Supplementary Material
Beautified Is Not Forged: A Two-Direction Measurement of Face-Forgery Detectors under Skin Smoothing

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Section, table and figure numbers without the prefix S refer to the paper. Names in small capitals such as BDRY38 are the result files of the released repository that hold the numbers of the section; pre-registration dates refer to its PRE_DECLARED log.

S1 Pre-registration and what was read when

Every bar quoted in the paper was written into a dated file before the corresponding model was trained or the corresponding test source was read. Celeb-DF-B and the DFD filter set were read once per model, after the checkpoint had been chosen on validation data. The Celeb-DF-v2 filter set was used to diagnose which filters cause the damage (Sec. S16), so it is a development source for every later design choice; the DFD filter set was built afterwards from a second filter library, and no per-filter result of it was inspected. Checkpoints are chosen on validation data only: FF++ validation macro-F1 (three-way, FF++ only), FF++ validation AUC (binary), or the mean of FF++ validation macro-F1 and balanced accuracy on held-out originals and renders of the two training services (three-way, full data). In the main paper the three-way model (OURS, full model) is the only 'Ours' row of the detection tables and OURS, forgery head only appears as the same-data control; this ordering was decided after both seeds of both models had been read; no model, checkpoint or number changed with that decision. A selection rule used before this paper (F1 of the beautified class on an imbalanced validation set) picked the epoch with the lowest specificity on untouched originals; it was replaced before the models of this paper were trained, and the model it selected appears only in Tables S40 and S41.

What pre-registration does not control is the choice of what to test. The commercial service of Sec. II was read for more than 60 of our checkpoints and all 17 published detectors; its strongest smoothing level was singled out after reading it; the first exposure analysis on it was post hoc; and the strength range of the scripted smoothing was set while the service's edit magnitudes were on disk. A calibration script also fitted two scaling constants of the scripted eye and jaw edits on 120 untouched originals of the test service; it changes no image and the binary arms do not use it. The arms of the smoothing chain (Sec. S17) were declared one at a time, each after the previous result had been read. The DF40 list, detectors and statistics were fixed before any detector was run on it; the scoring of the exposure arms and of CADDM on DF40 was added afterwards. The fusion pairs, rules and reading were fixed before any fused score was computed. In Celeb-DF-B, the trimming of the tinted bars that the app adds when it re-frames a video to portrait was introduced after the first extraction lost 57% of the treated faces; both extractions are kept, and the paper reports the trimmed one.

The results added in this version were declared in dated addenda before the corresponding model was trained or the source was read, with the following exceptions and amendments. The smoothed-genuine and smoothed-forgery test with skin-only forms (Sec. S9) and the entire-face-synthesis probe (Sec. S27) were declared as descriptive, with an expected direction and no bar. The dose-response rule (Sec. S11) and the readings of the forged-only arm (Sec. S15), the AR source (Sec. S6), the deployable threshold (Sec. S8) and the second seeds (Sec. S25) were declared before scoring. For the AR source the B-LFW sample was amended to 3,000 of 11,891 registered pairs before scoring, and the declaration's count of FairBeauty filters was corrected after reading the folders (eleven render folders hold eight distinct filters); the Spearman correlation of B-LFW with the commercial service was not declared. For the identity bootstrap (Sec. S12) the declared "merged" identity variant, which joins identities connected by swaps, was not usable on Celeb-DF (all 59 celebrities fall into four components holding every forged video) and is not reported.

S2 Cases near the class boundary

Table S1: Cases near the class boundary. "What changes" is the test the definition applies; it has not been measured (for example by face-recognition similarity or action-unit displacement) for any edit family. "Scored": whether a test set of the paper contains the case.

Case	Label	What changes	Scored
Camera processing, JPEG, video coding	neither	nothing about the person	all sets
Make-up worn at capture	neither	nothing after capture	incidental
Colour preset or filter	beautified	colour and tone	Celeb-DF-B, two filter sets
AR beauty filter (re-rendered face, identity kept)	beautified	skin, tone, smooth shape	FairBeauty, B-LFW
Skin smoothing, whitening (app or service, neural or not)	beautified	skin texture and tone	Celeb-DF-B, service
Face slimming, eye enlargement	beautified	smooth shape of own features	service, scripted
Generative "AI skin" that re-renders skin texture	beautified	skin texture, if shape and identity are kept	not scored
Relighting, virtual make-up transfer	beautified	appearance	not scored
Generative face restoration or super-resolution	boundary	texture; may alter identity cues	not scored
Face swap	forged	identity	FF++, Celeb-DF, DFD, DF40
Reenactment	forged	expression, motion	FF++, DF40
Lip-sync or dubbing	forged	speech, mouth motion	not scored
A part from another person	forged	identity of a part	part-level set
A part of the same person regenerated by inpainting	forged (boundary)	content of a part not captured	part-level set
Age or gender editing	forged (review if only texture changes)	identity cues not captured	not scored (DF40 face editing not available to us)
Appearance-only attribute edit (pale skin, make-up by a generator)	beautified	colour, tone and skin, identity and expression kept	not scored
Entire-face synthesis	forged	whole face not captured	DF40, six methods (probe)
Morphing of two identities	forged	identity	not scored
Beautified forgery; forgery of a beautified face	forged	identity or action	Celeb-DF-B, smoothed forgeries

DF40's dataset index lists four face-editing methods (e4e, StyleCLIP, CollabDiff, StarGAN-v2), but none of their folders was available to us, so editing is not scored; six entire-face-synthesis methods are scored as a probe (Sec. S27). The class definition reads only the forgery score $P(\text{fake})$; the beautified class of our three-way models is evaluated in Sec. S30.

S3 The published detectors

Table S2: Weights, training data and input of each published detector. On the video sources the twelve detectors trained on DeepfakeBench crops (the eight CNNs and the four CLIP detectors) are also scored on their own crop (dlib, five-point alignment, scale 1.3); SBI, UnivFD and NPR run on our crop; CADDM and LAA-Net re-align our crop with their own landmark or face models. On the retouching service and on DF40 every detector receives the same crop. UnivFD and NPR are general generated-image detectors, so the measurement covers fifteen face-forgery detectors and two general ones.

Detector	weights	training data	input
Xception, EfficientNet-B4, F3Net, SPSL, UCF, RECCE, CORE, SRM	DeepfakeBench release [1]	FF++ c23	256 px
SBI	authors' release (FFc23)	FF++ c23 reals, self-blended	380 px
Effort	authors' release	FF++ c23	224 px
Forensics Adapter	authors' release	FF++ c23	256 px, resized to 224 for CLIP
GenD (CLIP ViT-L/14)	authors' release	FF++ c23	224 px
DFD-HR	authors' release	FF++ c23	448 px
CADDM (EfficientNet-B4)	authors' release	FF++ raw	own 5-point alignment, 224 px
LAA-Net	authors' release (SBI-trained)	FF++ raw, self-blended	own crop rebuilt in our box, 384 px
UnivFD	authors' release (linear probe on CLIP)	ProGAN images (not faces)	224 px centre crop
NPR	authors' release	ProGAN images (not faces)	256 px

Table S3: Reported and reproduced. Published numbers are from DeepfakeBench (frame level, eight CNNs) and from the papers of the CLIP detectors. "DFB crop": our run of the released weights on DeepfakeBench crops (dlib frontal detector, 81 landmarks, five-point similarity alignment, scale 1.3, 256 px, 32 frames per video by linear spacing), built with the benchmark's own alignment function. "Our crop": landmark box with a 35% margin. Videos common to both frame sets (516 for Celeb-DF-v2, 332 for DFD). Our DFD set is a 332-video subset, so no published DFD number is comparable. Lower blocks: our full-data models (†trained on more data) on the DFB crop and on our crop, and their paired difference on our crop minus each CLIP detector on its DFB crop (95% video interval, first seeds).

	Celeb-DF-v2 frame AUC		Celeb-DF-v2 video AUC		DFD frame AUC	
Detector	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop	published	ours, DFB crop / our crop
Xception [2]	0.737	0.744 / 0.718	–	0.821 / 0.772	–	0.845 / 0.854
EfficientNet-B4 [3]	0.749	0.754 / 0.750	–	0.813 / 0.807	–	0.818 / 0.810
F3Net [4]	0.735	0.732 / 0.708	–	0.793 / 0.750	–	0.781 / 0.781
SPSL [5]	0.765	0.732 / 0.703	–	0.803 / 0.752	–	0.812 / 0.829
UCF [6]	0.753	0.775 / 0.744	–	0.841 / 0.788	–	0.818 / 0.822
RECCE [7]	0.732	0.744 / 0.688	–	0.827 / 0.731	–	0.839 / 0.833
CORE [8]	0.743	0.744 / 0.715	–	0.814 / 0.764	–	0.830 / 0.827
SRM [9]	0.755	0.763 / 0.717	–	0.844 / 0.778	–	0.837 / 0.830
Effort [10]	0.901	0.879 / 0.881	0.956	0.927 / 0.930	0.923	0.933 / 0.939
Forensics Adapter [11]	0.900	0.899 / 0.887	0.957	0.956 / 0.945	0.933	0.940 / 0.940
GenD [12]	–	0.900 / 0.891	0.960	0.960 / 0.954	–	0.932 / 0.934
DFD-HR [13]	0.910	0.886 / 0.879	0.960	0.951 / 0.943	0.953	0.942 / 0.943
OURS, forgery head only†	–	0.911 / 0.914	–	0.954 / 0.956	–	0.960 / 0.961
OURS, full model†	–	0.880 / 0.891	–	0.934 / 0.947	–	0.941 / 0.942
OURS, forgery head only (our crop) minus the detector (its DFB crop), 95% video interval						
Effort [10]		[+0.010, +0.059]		[+0.002, +0.055]		[+0.013, +0.045]
Forensics Adapter [11]		[-0.007, +0.034]		[-0.021, +0.020]		[+0.008, +0.035]
GenD [12]		[-0.005, +0.033]		[-0.023, +0.013]		[+0.017, +0.043]
DFD-HR [13]		[+0.010, +0.047]		[-0.013, +0.026]		[+0.008, +0.031]
OURS, full model (our crop) minus the detector (its DFB crop), 95% video interval						
Effort [10]		[-0.013, +0.038]		[-0.006, +0.044]		[-0.006, +0.025]
Forensics Adapter [11]		[-0.030, +0.013]		[-0.030, +0.010]		[-0.013, +0.017]
GenD [12]		[-0.028, +0.009]		[-0.031, +0.003]		[-0.004, +0.024]
DFD-HR [13]		[-0.014, +0.026]		[-0.022, +0.017]		[-0.012, +0.011]

Forensics Adapter reproduces its published Celeb-DF-v2 numbers on its own crop. Effort does not: our run gives 0.927 video AUC against 0.956 published, in line with an independent re-run of the released weights (0.932); comparisons of ours against Effort are therefore against the weaker reproduction. In the released training code of both, the checkpoint is saved when the metric on the configured test sets improves (Effort: `DeepfakeBench/training/trainer/trainer.py`, lines 377–390, called from `test_epoch`; Forensics Adapter: `trainer/trainer.py`, `save_best`, line 347); the papers do not state a selection rule. CADDM's released EfficientNet-B4 checkpoint gives a Celeb-DF-v2 video AUC of 0.937 on the authors' crop against 0.939 published; LAA-Net's released SBI checkpoint gives 0.943 against 0.954 on its own crop, which we rebuild inside our face box with a different face detector from the authors'. The lower blocks of Table S3 give our two full-data models on our crop and their paired differences against the CLIP detectors on the detectors' own crops (first seeds; the second seeds were not run through that script). Read as an audit (our models were trained on more data), OURS, forgery head only is above Effort's reproduction on both sets and above every CLIP detector on DFD frames, and on par with Forensics Adapter and GenD on Celeb-DF-v2; every interval of OURS, full model contains zero. At video level OURS, forgery head only is at 0.956 on Celeb-DF-v2 and 0.987 on DFD (second seed 0.963 and 0.978) and OURS, full model at 0.947 and 0.978 (0.946 and 0.981). A model trained on FF++ alone with our part-level forgeries does not reach OURS, forgery head only, so the margin of the full-data models comes from the external real data (FFHQ stills and commercial renders), not from the synthesis.

S4 Comparison with prior evaluations

Table S4: Our evaluation against prior evaluations of beautification and forgery detection. Detectors: number scored with released weights. Both: accusation of beautified genuine faces and escape of beautified forgeries read on the same detector. Pairs: beautified face read against its own untouched original. Compr.: re-encoding controlled. Cause: training intervention with everything else fixed. Thr.: a threshold rule tested. Rel.: sources and scorer released. –: not done or not applicable.

Evaluation	Det.	Both	Pairs	Compr.	Cause	Thr.	Rel.
Libourel et al. [14], 2024	3	–	–	✓	–	–	✓
Saeed et al. [15], 2025	6	–	–	–	partial	–	–
Concas et al. [16], 2025	2	✓	–	–	–	–	–
BIAS-ID [17], 2026	–	✓	–	–	2 cells	–	✓
LAION-Mobile [18], 2026	12	–	–	–	–	✓	✓
This paper	**17**	✓	✓	✓	4 cells	✓	✓

S5 Both errors in full

Table S5: All three beautification sources with intervals (frame level). Celeb-DF-B as in Table II; Celeb-DF-v2 and DFD: eleven and sixteen held-out filters on the full frame, ΔMISS and ΔFA with 95% video-cluster intervals.

	Celeb-DF-B (app presets)				Celeb-DF-v2 + 11 held-out filters				DFD + 16 fresh filters			
Detector	AUC	AUC_b	$\Delta MISS_{beauty}$	FA_b	AUC	AUC_f	ΔMISS	ΔFA	AUC	AUC_f	ΔMISS	ΔFA
Xception [2]	0.811	0.737	+10.9 [7.4, 14.1]	5.7	0.718	0.693	+1.1 [-1.0, 3.1]	+1.5 [-0.2, 3.2]	0.854	0.834	+2.0 [-0.6, 5.4]	+2.0 [0.2, 4.3]
EfficientNet-B4 [3]	0.811	0.762	+3.4 [0.9, 5.9]	7.8	0.750	0.725	+2.5 [0.2, 4.8]	+0.8 [-0.3, 2.6]	0.810	0.781	+6.3 [3.6, 9.2]	-0.5 [-1.9, 1.7]
F3Net [4]	0.798	0.731	+10.9 [7.4, 14.8]	5.4	0.708	0.688	-0.6 [-2.7, 1.6]	+1.3 [-0.8, 3.8]	0.781	0.774	-8.5 [-11.5, -5.5]	+3.6 [2.1, 5.7]
SPSL [5]	0.781	0.727	+12.4 [9.3, 15.5]	3.4	0.703	0.683	+3.1 [1.5, 4.9]	-0.2 [-1.5, 1.4]	0.829	0.837	+4.7 [3.2, 6.7]	-1.8 [-2.4, -0.9]
UCF [6]	0.841	0.777	+11.3 [7.6, 15.4]	5.7	0.744	0.697	+3.1 [1.0, 5.5]	+0.9 [-0.6, 2.3]	0.822	0.804	+5.0 [1.9, 8.0]	+0.3 [-1.6, 2.5]
RECCE [7]	0.785	0.723	+13.8 [10.7, 17.1]	4.1	0.688	0.661	+0.1 [-1.9, 2.0]	+0.9 [-0.8, 2.4]	0.833	0.808	+7.3 [4.6, 10.7]	-0.2 [-1.7, 1.3]
CORE [8]	0.821	0.752	+13.5 [10.2, 16.6]	4.0	0.715	0.693	-0.0 [-1.5, 1.9]	+0.4 [-0.8, 2.4]	0.827	0.821	-5.4 [-8.8, -2.9]	+1.6 [-0.3, 3.7]
SRM [9]	0.789	0.733	+15.3 [12.1, 18.7]	2.8	0.717	0.712	-0.1 [-1.4, 1.6]	+0.7 [-0.5, 2.1]	0.830	0.822	+5.8 [2.4, 9.7]	+0.2 [-1.2, 1.9]
SBI [19]	0.875	0.752	+21.8 [17.3, 26.1]	1.8	0.817	0.776	+8.2 [5.5, 10.4]	-0.6 [-1.5, 0.5]	0.829	0.810	+4.5 [2.7, 6.3]	+0.4 [-0.7, 1.4]
UnivFD‡ [20]	0.653	0.624	-7.2 [-10.2, -4.9]	7.6	0.662	0.622	-7.2 [-10.6, -4.2]	+7.8 [4.1, 11.7]	0.840	0.801	-10.2 [-13.5, -6.8]	+11.6 [8.0, 15.7]
NPR‡ [21]	0.503	0.498	-0.7 [-2.2, 0.4]	3.4	0.522	0.497	-38.0 [-43.2, -32.2]	+36.0 [28.7, 43.6]	0.473	0.516	-13.7 [-20.2, -8.4]	+9.4 [2.4, 15.3]
Effort [10]	0.909	0.807	+11.0 [7.7, 14.5]	5.4	0.881	0.813	+23.3 [19.9, 26.3]	-2.8 [-3.7, -1.7]	0.939	0.905	+15.1 [11.1, 18.6]	-2.3 [-3.1, -1.4]
Forensics Adapter [11]	0.909	0.828	+15.2 [11.9, 18.6]	7.9	0.887	0.814	+11.9 [8.6, 14.7]	+2.5 [0.3, 5.7]	0.940	0.924	+8.0 [5.6, 11.2]	-0.6 [-1.6, 0.4]
GenD [12]	0.924	0.807	+12.0 [8.2, 15.6]	9.9	0.891	0.856	-10.1 [-13.1, -7.1]	+9.9 [7.4, 13.1]	0.934	0.925	+3.1 [0.7, 5.8]	+1.2 [0.1, 2.2]
DFD-HR [13]	0.938	0.838	+3.6 [0.6, 6.8]	14.6	0.879	0.824	+8.0 [5.3, 10.5]	+1.6 [0.4, 3.8]	0.943	0.928	+9.1 [6.8, 11.8]	-2.0 [-2.9, -1.2]
CADDM [22]	0.886	0.762	-2.9 [-5.8, 0.1]	13.8	0.839	0.796	+3.2 [0.7, 5.7]	+3.0 [1.4, 4.7]	0.845	0.818	+3.8 [2.2, 5.9]	+0.8 [-0.4, 1.9]
LAA-Net [23]	0.889	0.698	+14.7 [10.9, 18.6]	6.0	0.851	0.799	+10.2 [7.4, 12.8]	-0.1 [-1.4, 1.1]	0.810	0.783	+6.2 [4.2, 7.7]	+0.0 [-0.9, 1.0]
FF++-only three-way	0.938	0.872	+7.5 [4.8, 10.5]	7.8	0.898	0.851	-3.5 [-6.1, -1.0]	+7.2 [4.8, 10.9]	0.943	0.932	-0.3 [-2.5, 2.3]	+3.4 [2.1, 5.1]
Ours, forgery head only	0.949	0.896	+7.0 [4.7, 9.0]	7.4	0.914	0.883	+9.8 [7.4, 12.5]	-0.4 [-1.8, 0.9]	0.961	0.952	+4.2 [2.3, 6.5]	+0.2 [-0.9, 1.4]
second seed	0.955	0.904	+5.7 [3.5, 7.8]	8.3	0.923	0.895	+6.2 [3.4, 8.4]	-0.1 [-1.1, 1.0]	0.951	0.938	+3.8 [1.4, 6.0]	+1.1 [-0.2, 2.8]
Ours, full model	0.938	0.866	+6.7 [4.0, 9.5]	9.5	0.891	0.873	+7.3 [4.3, 9.9]	-0.4 [-1.4, 0.6]	0.942	0.931	+8.9 [6.2, 12.1]	-1.0 [-2.4, 0.1]
second seed	0.941	0.883	+8.3 [5.6, 11.0]	6.7	0.887	0.875	+2.0 [-1.1, 4.5]	+0.4 [-1.1, 2.4]	0.945	0.931	+4.4 [2.2, 7.7]	+0.3 [-0.8, 1.7]
Ours, full model, sym. view	0.941	0.868	+10.0 [7.0, 13.1]	9.2	0.899	0.881	+6.7 [3.9, 9.6]	-0.6 [-1.6, 0.7]	0.947	0.934	+8.5 [5.7, 12.0]	-0.9 [-2.2, -0.0]
second seed	0.945	0.888	+3.1 [0.6, 5.8]	8.7	0.901	0.874	-0.7 [-3.1, 2.0]	+2.8 [0.6, 4.7]	0.954	0.940	+2.6 [0.3, 4.9]	+2.9 [1.6, 4.3]

Table S6: Celeb-DF-B: re-encoding separated from the preset on both errors, frame and video level. Video AUC: mean frame score per video. FA_c, FA_b: false alarms on re-encoded and beautified genuine frames at the threshold of 5% FPR on untreated genuine frames; $\Delta FA_{beauty} = FA_b - FA_c$ with a 95% interval over 1,000 draws (threshold re-estimated in each) that resample videos (as in the earlier version) or identities (58 units; the draws of Sec. S12).

	frame AUC			video AUC			false alarms (%)			
Detector	untreated	re-encoded	beautified	untreated	re-encoded	beautified	FA_c	FA_b	ΔFA_{beauty} [video CI]	[identity CI]
Xception [2]	0.811	0.795	0.737	0.869	0.855	0.774	7.4	5.7	-1.7 [-3.0, -0.4]	–
EfficientNet-B4 [3]	0.811	0.798	0.762	0.870	0.849	0.816	6.1	7.8	+1.7 [+0.2, +3.6]	–
F3Net [4]	0.798	0.782	0.731	0.856	0.834	0.769	7.3	5.4	-1.9 [-3.5, -0.3]	–
SPSL [5]	0.781	0.769	0.727	0.849	0.832	0.775	5.8	3.4	-2.4 [-3.4, -1.6]	–
UCF [6]	0.841	0.829	0.777	0.897	0.883	0.818	5.6	5.7	+0.2 [-1.4, +1.7]	–
RECCE [7]	0.785	0.765	0.723	0.847	0.824	0.766	7.8	4.1	-3.6 [-5.0, -2.3]	–
CORE [8]	0.821	0.808	0.752	0.880	0.867	0.797	6.2	4.0	-2.2 [-3.3, -0.9]	–
SRM [9]	0.789	0.769	0.733	0.857	0.842	0.788	6.2	2.8	-3.4 [-4.3, -2.5]	–
SBI [19]	0.875	0.861	0.752	0.922	0.906	0.799	2.4	1.8	-0.6 [-1.6, +0.3]	–
UnivFD‡ [20]	0.653	0.637	0.624	0.702	0.681	0.658	3.1	7.6	+4.5 [+3.1, +6.7]	–
NPR‡ [21]	0.503	0.503	0.498	0.501	0.504	0.494	2.5	3.4	+0.9 [-0.3, +2.2]	–
Effort [10]	0.909	0.882	0.807	0.952	0.933	0.852	3.3	5.4	+2.1 [+0.2, +4.1]	[-0.1, +4.0]
Forensics Adapter [11]	0.909	0.894	0.828	0.961	0.949	0.888	7.8	7.9	+0.1 [-1.6, +2.3]	[-1.6, +2.4]
GenD [12]	0.924	0.898	0.807	0.971	0.953	0.854	6.3	9.9	+3.6 [+1.2, +6.6]	[+0.9, +6.4]
DFD-HR [13]	0.938	0.915	0.838	0.979	0.963	0.890	4.8	14.6	+9.8 [+6.5, +13.1]	[+6.4, +13.2]
CADDM [22]	0.886	0.850	0.762	0.965	0.933	0.832	3.9	13.8	+9.9 [+7.7, +12.1]	–
LAA-Net [23]	0.889	0.870	0.698	0.960	0.944	0.743	2.7	6.0	+3.3 [+1.6, +5.1]	–
OURS, forgery head only	0.949	0.934	0.896	0.982	0.974	0.948	5.0	7.4	+2.3 [+1.2, +3.8]	[+1.2, +3.7]
second seed	0.955	0.939	0.904	0.986	0.978	0.952	5.1	8.3	+3.2 [+1.7, +4.8]	[+1.7, +4.8]
OURS, full model	0.938	0.918	0.866	0.979	0.966	0.920	4.6	9.5	+4.9 [+2.8, +7.5]	[+2.6, +7.6]
second seed	0.941	0.926	0.883	0.984	0.977	0.941	4.0	6.7	+2.7 [+1.3, +4.1]	[+1.2, +4.2]
OURS, full model, sym. view	0.941	0.925	0.868	0.981	0.974	0.918	4.3	9.2	+4.9 [+2.7, +7.3]	[+2.4, +7.6]
second seed	0.945	0.934	0.888	0.981	0.977	0.937	3.8	8.7	+4.9 [+3.2, +6.8]	[+3.4, +6.8]
FF++-only three-way	0.938	0.915	0.872	0.973	0.958	0.920	6.0	7.8	+1.8 [-0.1, +4.2]	[-0.2, +4.4]
P0	0.927	0.901	0.860	0.961	0.940	0.909	6.7	5.1	-1.6 [-3.2, +0.1]	[-3.3, +0.4]
P2-sym	0.920	0.900	0.866	0.961	0.942	0.916	6.7	5.5	-1.2 [-2.7, +0.5]	[-2.5, +0.4]
second seed	0.933	0.914	0.888	0.963	0.947	0.928	6.6	6.2	-0.3 [-1.8, +1.3]	[-1.9, +1.5]

Table S7: Celeb-DF-B, paired differences of the frame AUC under the presets (AUC_b), our model minus each CLIP detector, with 95% intervals over identical draws that resample videos (first line of each model) or identities (second line; Sec. S12).

Model	resampling	Effort	Forensics Adapter	GenD	DFD-HR
OURS, forgery head only	video	+0.088 [+0.060, +0.119]	+0.068 [+0.042, +0.094]	+0.088 [+0.060, +0.118]	+0.058 [+0.034, +0.083]
	identity	[+0.065, +0.114]	[+0.048, +0.087]	[+0.066, +0.110]	[+0.039, +0.076]
second seed	video	+0.097 [+0.070, +0.125]	+0.076 [+0.051, +0.100]	+0.097 [+0.072, +0.126]	+0.066 [+0.044, +0.090]
	identity	[+0.075, +0.120]	[+0.057, +0.094]	[+0.075, +0.118]	[+0.048, +0.084]
OURS, full model	video	+0.059 [+0.030, +0.089]	+0.038 [+0.013, +0.064]	+0.059 [+0.032, +0.086]	+0.028 [+0.005, +0.052]
	identity	[+0.033, +0.086]	[+0.016, +0.059]	[+0.036, +0.081]	[+0.008, +0.048]
second seed	video	+0.075 [+0.050, +0.103]	+0.055 [+0.031, +0.079]	+0.075 [+0.051, +0.104]	+0.045 [+0.022, +0.068]
	identity	[+0.052, +0.101]	[+0.034, +0.074]	[+0.052, +0.097]	[+0.026, +0.063]
OURS, full model, sym. view	video	+0.061 [+0.032, +0.091]	+0.040 [+0.015, +0.065]	+0.061 [+0.035, +0.088]	+0.030 [+0.007, +0.054]
	identity	[+0.035, +0.088]	[+0.018, +0.062]	[+0.037, +0.083]	[+0.011, +0.048]
second seed	video	+0.081 [+0.055, +0.107]	+0.060 [+0.036, +0.084]	+0.081 [+0.056, +0.108]	+0.050 [+0.029, +0.072]
	identity	[+0.057, +0.105]	[+0.039, +0.081]	[+0.057, +0.103]	[+0.031, +0.069]
FF++-only three-way	video	+0.065 [+0.041, +0.092]	+0.044 [+0.019, +0.068]	+0.065 [+0.041, +0.090]	+0.035 [+0.014, +0.055]
	identity	[+0.041, +0.091]	[+0.022, +0.066]	[+0.047, +0.084]	[+0.017, +0.051]
P0	video	+0.053 [+0.026, +0.081]	+0.032 [+0.007, +0.058]	+0.053 [+0.028, +0.078]	+0.022 [+0.002, +0.042]
	identity	[+0.026, +0.079]	[+0.009, +0.054]	[+0.036, +0.070]	[+0.006, +0.038]
P2-sym	video	+0.059 [+0.030, +0.088]	+0.038 [+0.011, +0.064]	+0.058 [+0.033, +0.088]	+0.028 [+0.006, +0.050]
	identity	[+0.033, +0.084]	[+0.016, +0.061]	[+0.037, +0.080]	[+0.008, +0.048]
second seed	video	+0.081 [+0.055, +0.109]	+0.060 [+0.034, +0.085]	+0.081 [+0.056, +0.111]	+0.051 [+0.029, +0.074]
	identity	[+0.057, +0.107]	[+0.038, +0.081]	[+0.057, +0.104]	[+0.027, +0.072]

The data hold 232 videos per condition; 231 have at least one detected face in every condition except the beautified genuine one (230). Re-encoding alone lowers the frame AUC of every detector by at most 0.036, and the presets lower it further by 0.006–0.172 (Table S6); for the four CLIP detectors the loss beyond re-encoding is 0.066–0.091, for OURS, forgery head only 0.038 and 0.035 and for OURS, full model 0.052 and 0.043. On the false-alarm side, beyond re-encoding, the presets add 9.8 points for DFD-HR [6.5, 13.1], 9.9 for CADDM [7.7, 12.1] and 3.6 for GenD [1.2, 6.6], against 2.1 for Effort and

0.1 for Forensics Adapter; OURS, forgery head only adds 2.3 [1.2, 3.8] (second seed 3.2) and OURS, full model 4.9 [2.8, 7.5] (2.7). At video level CADDM goes from 0.965 untreated to 0.933 re-encoded and 0.832 beautified; Libourel et al. report 0.91 to 0.76 for it on their own frames. Two confounds remain. The app re-frames treated videos to portrait with tinted bars, so face size and resampling differ between untreated and treated frames, and the re-encoded control reproduces the codec but not the re-framing; we did not build a face-size-matched subset. The subtraction assumes that the effects of re-encoding and preset add.

Table S8: The same videos, frames and per-video filter assignment with the filter applied to the face crop or to the full frame before the crop (frame AUC). A filter with a vignette is centred on the face in the first case. The paper reports the full-frame version.

	Celeb-DF-v2, 11 filters			DFD, 16 filters		
Detector	AUC	AUC_f crop	AUC_f full frame	AUC	AUC_f crop	AUC_f full frame
Xception [2]	0.718	0.673	0.693	0.854	0.823	0.834
EfficientNet-B4 [3]	0.750	0.698	0.725	0.810	0.777	0.781
F3Net [4]	0.708	0.663	0.688	0.781	0.769	0.774
SPSL [5]	0.703	0.664	0.683	0.829	0.834	0.837
UCF [6]	0.744	0.669	0.697	0.822	0.793	0.804
RECCE [7]	0.688	0.636	0.661	0.833	0.795	0.808
CORE [8]	0.715	0.669	0.693	0.827	0.813	0.821
SRM [9]	0.717	0.696	0.712	0.830	0.811	0.822
SBI [19]	0.817	0.729	0.776	0.829	0.803	0.810
UnivFD‡ [20]	0.662	0.610	0.622	0.840	0.793	0.801
NPR‡ [21]	0.522	0.502	0.497	0.473	0.541	0.516
Effort [10]	0.881	0.788	0.813	0.939	0.904	0.905
Forensics Adapter [11]	0.887	0.790	0.814	0.940	0.907	0.924
GenD [12]	0.891	0.818	0.856	0.934	0.913	0.925
DFD-HR [13]	0.879	0.781	0.824	0.943	0.921	0.928
OURS, forgery head only	0.914	0.863	0.883	0.961	0.949	0.952
OURS, full model	0.891	0.849	0.873	0.942	0.927	0.931
FF++-only three-way	0.898	0.797	0.851	0.943	0.925	0.932

S6 Instagram AR filters

Table S9: Instagram AR filter renders against their own originals [24]. Threshold: 95th percentile of each source's originals, re-estimated in each of 1,000 draws (FairBeauty: pairs; B-LFW: identities). "Accuses": lower interval bound above 5%. JPEG q90: FairBeauty with both originals and renders re-compressed. †Trained on more data than the published detectors.

	FairBeauty (2,927 pairs)				B-LFW (3,000 pairs, 2,016 identities)		
Detector	accused (%) [95% CI]	AUROC	above own orig. (%)	accused, JPEG q90	accused (%) [95% CI]	AUROC	above own orig. (%)
Xception [2]	24.3 [21.7, 28.3]	0.793	85.7	18.9	53.3 [49.4, 57.2]	0.876	95.1
EfficientNet-B4 [3]	22.0 [19.6, 24.2]	0.751	83.1	18.1	39.9 [36.0, 43.8]	0.859	96.3
F3Net [4]	26.1 [23.3, 29.6]	0.751	83.3	17.1	49.0 [45.9, 53.4]	0.869	95.3
SPSL [5]	10.7 [9.1, 13.2]	0.691	77.2	10.3	12.0 [10.5, 14.3]	0.684	82.1
UCF [6]	37.3 [32.8, 41.1]	0.820	90.4	29.3	71.5 [68.5, 73.7]	0.926	97.8
RECCE [7]	41.5 [38.8, 44.6]	0.815	89.8	28.7	78.4 [75.2, 80.5]	0.941	97.9
CORE [8]	25.2 [23.4, 28.1]	0.776	86.5	18.4	58.8 [56.5, 62.5]	0.885	95.3
SRM [9]	26.8 [24.1, 29.7]	0.787	85.6	17.1	58.1 [55.1, 61.1]	0.895	96.7
SBI [19]	22.8 [20.2, 24.9]	0.671	70.0	20.0	35.4 [33.3, 38.2]	0.745	81.9
UnivFD‡ [20]	63.3 [59.5, 66.5]	0.900	97.2	54.7	48.6 [45.9, 52.3]	0.855	96.6
NPR‡ [21]	70.4 [63.8, 78.5]	0.960	97.0	4.5	0.0 [0.0, 0.0]	0.468	2.8
Effort [10]	67.8 [64.2, 70.5]	0.905	96.0	66.1	74.7 [71.8, 76.6]	0.942	98.6
Forensics Adapter [11]	63.9 [61.8, 66.4]	0.877	95.3	66.9	76.9 [75.1, 79.0]	0.925	98.9
GenD [12]	75.6 [73.4, 78.8]	0.949	97.8	74.0	96.6 [95.7, 97.8]	0.992	99.9
DFD-HR [13]	85.9 [83.8, 87.8]	0.966	98.8	87.1	95.3 [94.2, 96.2]	0.988	99.7
CADDM [22]	15.4 [13.1, 17.9]	0.700	76.4	10.4	35.8 [32.9, 39.9]	0.807	86.2
LAA-Net [23]	27.6 [24.7, 30.1]	0.667	68.9	20.3	48.4 [45.5, 50.4]	0.779	80.7
OURS, forgery head only†	79.0 [76.2, 81.4]	0.954	98.6	73.4	89.0 [87.5, 91.6]	0.978	99.4
second seed†	82.1 [79.9, 84.2]	0.961	98.9	73.2	90.3 [88.3, 92.6]	0.980	99.7
OURS, full model†	75.1 [72.7, 77.6]	0.945	98.4	64.3	86.7 [83.5, 88.9]	0.975	99.4
OURS, full model, sym. view†	77.4 [75.5, 79.6]	0.950	98.6	67.3	93.6 [91.7, 95.0]	0.988	99.8
second seed†	75.6 [72.8, 77.7]	0.943	97.8	71.8	94.3 [92.9, 95.3]	0.986	99.8

FairBeauty's eleven render folders hold eight distinct filters (folders 8–10 repeat three of them; folder 10 has one pair). B-LFW renders are 112 px; its originals were registered onto the render frame. All seventeen published detectors accuse FairBeauty renders and sixteen accuse B-LFW renders; NPR's scores saturate on B-LFW and rank the originals above the renders (2.8% of pairs). After re-compressing both sides, only NPR loses more than half of its accused share (ratio 0.064); the other detectors keep 0.64–1.05 and our models 0.86–0.95. The AR engine's capture path is still confounded with the beautification, so these numbers are an upper bound on the effect of the beautification itself. Over the seventeen, the Spearman correlation of the accused share with the commercial Smoothing-90 share is 0.15 ($p = 0.58$) for FairBeauty and 0.59 ($p = 0.013$; not declared) for B-LFW. OURS, forgery head only minus GenD is $+3.4$ [-0.6, $+6.2$] on FairBeauty and -7.5 [-9.3, -5.1] on B-LFW, minus DFD-HR -6.8 and -6.2 (first seed); OURS, full model minus GenD is -0.5 [-3.9, $+2.5$] and -9.9 [-13.2, -7.8], minus DFD-HR -10.7 and -8.6.

S7 The operation head on Instagram AR filters

The four operation scores of the three-way model were read on the FairBeauty pairs (2,919 of the 2,927 pairs measurable). The presence of an operation in a render is measured on the pair itself (eye enlargement, jaw change, whitening and smoothing from landmarks and skin statistics, with half of the Smoothing-30 median effect of the commercial service as the rule; "strict" uses the full median). Table S10 gives the measured presence per filter and the head's recall and precision per operation. For the model of the paper the recall is 0.003 (eye), 0.032 (jaw), 0.176 (whitening) and 0.243 (smoothing); the exact four-bit match is 1.9%. The multi-operation arm (whose pre-declared AR endpoint failed) and a model trained with the AR pairs are listed for reference; the latter learns the capture pipeline (Sec. S32).

Table S10: Operation head under Instagram AR filters (FB_OPS). Top: measured presence (%) per filter group. Bottom: share of renders and of originals on which the head fires, recall and precision per operation and model.

Group	n	eye	jaw	white	smooth	strict eye / jaw / white / smooth
all	2919	80.5	8.5	39.1	96.6	76.7 / 6.8 / 35.3 / 93.5
held_out_filters	881	80.0	9.8	20.3	98.0	76.2 / 7.6 / 16.6 / 94.8
training_filters	2038	80.8	8.0	47.2	96.0	77.0 / 6.5 / 43.4 / 92.9
Baby_cute_face	294	75.9	8.8	54.4	99.3	70.7 / 6.8 / 43.5 / 98.3
Caramel Macchiato	295	90.2	10.8	57.6	100.0	87.1 / 8.5 / 55.9 / 99.0
Cute baby face	298	73.5	7.4	88.6	99.0	68.8 / 6.4 / 85.2 / 98.7
Just Baby	293	82.3	9.2	3.1	98.0	79.2 / 6.5 / 2.7 / 97.6
Pretty	566	81.4	7.4	29.7	97.0	76.5 / 6.4 / 25.1 / 93.8
Shiny foxy	295	86.1	6.1	57.3	96.9	83.4 / 4.4 / 53.6 / 95.6
big city life	294	82.0	11.2	3.4	96.6	78.6 / 9.5 / 3.4 / 88.4
hary beauty	584	76.4	8.4	32.5	91.1	73.3 / 6.7 / 28.4 / 84.6

Model	group	op.	head on render (%)	head on original (%)	recall	precision
OURS (three-way)	all	eye	0.4	3.2	0.003	0.667
OURS (three-way)	all	jaw	2.1	6.8	0.032	0.131
OURS (three-way)	all	white	12.7	18.2	0.176	0.543
OURS (three-way)	all	smooth	24.8	6.9	0.243	0.946
OURS (three-way)	held_out_filters	eye	0.6	3.1	0.004	0.6
OURS (three-way)	held_out_filters	jaw	2.7	7.0	0.047	0.167
OURS (three-way)	held_out_filters	white	13.3	16.0	0.156	0.239
OURS (three-way)	held_out_filters	smooth	21.3	6.8	0.212	0.973
OURS (three-way)	training_filters	eye	0.3	3.2	0.003	0.714
OURS (three-way)	training_filters	jaw	1.8	6.7	0.025	0.108
OURS (three-way)	training_filters	white	12.4	19.1	0.18	0.684
OURS (three-way)	training_filters	smooth	26.3	6.9	0.257	0.937
multi-operation views	all	eye	3.7	5.9	0.036	0.778
multi-operation views	all	jaw	10.0	21.9	0.116	0.099
multi-operation views	all	white	19.7	19.2	0.265	0.525
multi-operation views	all	smooth	52.7	14.2	0.529	0.969
multi-operation views	held_out_filters	eye	6.0	6.6	0.061	0.811
multi-operation views	held_out_filters	jaw	16.7	23.5	0.116	0.068
multi-operation views	held_out_filters	white	25.3	17.9	0.285	0.229
multi-operation views	held_out_filters	smooth	62.2	14.6	0.628	0.989
multi-operation views	training_filters	eye	2.7	5.6	0.025	0.745
multi-operation views	training_filters	jaw	7.2	21.2	0.117	0.13
multi-operation views	training_filters	white	17.3	19.7	0.261	0.713
multi-operation views	training_filters	smooth	48.6	14.0	0.485	0.959
trained with AR pairs	all	eye	52.1	0.0	0.512	0.792
trained with AR pairs	all	jaw	40.5	0.5	0.365	0.077
trained with AR pairs	all	white	91.1	0.1	0.921	0.395
trained with AR pairs	all	smooth	99.9	0.1	0.999	0.966
trained with AR pairs	held_out_filters	eye	44.7	0.0	0.444	0.794
trained with AR pairs	held_out_filters	jaw	35.1	0.7	0.314	0.087
trained with AR pairs	held_out_filters	white	88.1	0.0	0.922	0.213
trained with AR pairs	held_out_filters	smooth	99.9	0.2	0.999	0.98
trained with AR pairs	training_filters	eye	55.3	0.0	0.541	0.791
trained with AR pairs	training_filters	jaw	42.8	0.4	0.393	0.073
trained with AR pairs	training_filters	white	92.3	0.1	0.921	0.47
trained with AR pairs	training_filters	smooth	100.0	0.0	0.999	0.96

S8 A deployable threshold

Table S11: One threshold per detector: the 95th percentile of $P(\text{fake})$ on the 530 genuine FF++ c23 validation frames (133 videos), with its 95% interval over validation videos. Cells: rate at this threshold / rate at the source's own 5%-FPR threshold (%). FA on genuine conditions, MISS on the beautified Celeb-DF-B forgeries. FF++ validation also selected our checkpoints, so our thresholds are, if anything, favourable to us.

Detector	threshold [CI]	CDF-B untr.	CDF-B beaut.	CDF-B beaut. forged (miss)	DFD filt.	service orig.	Smoothing-90	FairBeauty	B-LFW
Xception [2]	0.84 [0.79, 0.88]	6.3 / 5.0	6.9 / 5.7	68.3 / 71.2	11.3 / 7.0	10.7 / 5.0	65.0 / 51.6	21.2 / 24.3	37.4 / 53.3
EfficientNet-B4 [3]	0.86 [0.80, 0.94]	5.2 / 5.0	8.1 / 7.8	67.1 / 67.7	17.6 / 4.6	19.2 / 5.0	63.8 / 36.3	44.6 / 22.0	82.6 / 39.9
F3Net [4]	0.79 [0.68, 0.89]	6.9 / 5.0	7.2 / 5.4	72.8 / 77.7	22.4 / 8.6	12.7 / 5.0	69.0 / 50.6	36.7 / 26.1	63.6 / 49.0
SPSL [5]	0.95 [0.92, 0.97]	3.8 / 5.0	2.6 / 3.4	84.9 / 82.0	11.3 / 3.2	6.8 / 5.0	56.3 / 51.2	15.9 / 10.7	48.1 / 12.0
UCF [6]	0.99 [0.95, 1.00]	2.4 / 5.0	3.2 / 5.7	76.2 / 67.1	9.7 / 5.3	7.0 / 5.0	48.4 / 41.4	29.5 / 37.3	65.1 / 71.5
RECCE [7]	0.87 [0.78, 0.94]	8.8 / 5.0	6.9 / 4.1	74.0 / 81.8	17.9 / 4.8	12.7 / 5.0	60.0 / 36.8	44.7 / 41.5	83.2 / 78.4
CORE [8]	0.76 [0.69, 0.83]	8.2 / 5.0	6.4 / 4.0	68.9 / 75.9	19.7 / 6.6	10.6 / 5.0	66.7 / 54.8	33.5 / 25.2	73.3 / 58.8
SRM [9]	0.65 [0.64, 0.66]	4.4 / 5.0	2.6 / 2.8	88.5 / 88.0	10.4 / 5.2	4.8 / 5.0	62.0 / 62.8	24.7 / 26.8	53.6 / 58.1
SBI [19]	0.29 [0.18, 0.53]	38.5 / 5.0	23.2 / 1.8	40.1 / 79.8	41.0 / 5.4	45.3 / 5.0	93.3 / 37.3	82.1 / 22.8	87.5 / 35.4
UnivFD‡ [20]	0.20 [0.15, 0.35]	3.0 / 5.0	4.8 / 7.6	84.3 / 79.8	6.1 / 16.6	4.3 / 5.0	15.2 / 16.6	61.4 / 63.3	54.8 / 48.6
NPR‡ [21]	1.00 [0.85, 1.00]	1.4 / 5.0	0.1 / 3.4	99.8 / 96.4	10.0 / 14.4	n/a	n/a	7.0 / 70.4	97.3 / 0.0
Effort [10]	0.76 [0.66, 0.85]	6.6 / 5.0	7.4 / 5.4	47.3 / 51.4	4.7 / 2.7	1.4 / 5.0	31.1 / 43.2	75.1 / 67.8	80.2 / 74.7
Forensics Adapter [11]	0.76 [0.65, 0.84]	7.8 / 5.0	11.2 / 7.9	40.6 / 46.5	3.7 / 4.4	3.1 / 5.0	32.5 / 39.7	59.1 / 63.9	69.3 / 76.9
GenD [12]	0.84 [0.71, 0.89]	5.7 / 5.0	11.2 / 9.9	45.3 / 48.1	6.4 / 6.3	4.8 / 5.0	67.4 / 68.7	87.3 / 75.6	96.9 / 96.6
DFD-HR [13]	0.64 [0.58, 0.71]	4.5 / 5.0	13.5 / 14.6	36.4 / 34.7	7.3 / 3.0	2.9 / 5.0	56.2 / 64.9	93.4 / 85.9	97.2 / 95.3
CADDM [22]	0.89 [0.67, 0.98]	3.9 / 5.0	11.1 / 13.8	57.9 / 54.2	12.0 / 5.8	32.9 / 5.0	42.3 / 6.3	66.8 / 15.4	79.2 / 35.8
LAA-Net [23]	0.76 [0.56, 0.88]	6.9 / 5.0	7.7 / 6.0	69.0 / 73.3	13.2 / 5.0	30.9 / 5.0	81.0 / 17.2	52.3 / 27.6	50.8 / 48.4
Ours, forgery head only†	0.98 [0.93, 1.00]	7.7 / 5.0	10.5 / 7.4	30.3 / 36.6	5.7 / 5.2	0.1 / 5.0	0.2 / 9.1	82.3 / 79.0	93.2 / 89.0
second seed†	0.85 [0.60, 0.97]	11.6 / 4.9	17.3 / 8.3	19.2 / 30.9	12.5 / 6.1	0.2 / 5.0	0.1 / 6.5	91.2 / 82.1	97.6 / 90.3
Ours, full model†	0.65 [0.51, 0.87]	10.4 / 5.0	16.3 / 9.5	28.5 / 40.7	8.9 / 4.0	0.4 / 4.9	0.5 / 8.6	90.6 / 75.1	98.0 / 86.7
second seed†	0.94 [0.57, 0.97]	4.4 / 5.0	5.7 / 6.7	43.6 / 41.3	5.8 / 5.3	0.1 / 5.0	0.1 / 12.5	–	–
Ours, full model, sym. view†	0.73 [0.49, 0.90]	9.7 / 5.0	14.5 / 9.2	31.8 / 42.5	8.9 / 4.0	0.3 / 5.0	0.1 / 8.1	91.5 / 77.4	99.2 / 93.6
second seed†	0.76 [0.45, 0.92]	7.5 / 5.0	12.5 / 8.7	28.3 / 34.1	8.7 / 7.9	0.1 / 5.0	0.1 / 17.8	88.0 / 75.6	97.4 / 94.3

The rate on untreated genuine faces is the calibration drift of each detector: SBI calls 38.5% of untreated Celeb-DF-B genuine frames and 45.3% of the service's FFHQ originals fake; Ours, forgery head only (first seed) 7.7% and 0.1%, Ours, full model 10.4% and 0.4%. On the AR renders every detector except NPR accuses about as much or more at the deployable threshold than at the per-source one. The Spearman correlation between the detector rankings under the two rules is 0.29 ($p = 0.25$) for Smoothing-90, 0.37 ($p = 0.15$) for FairBeauty, 0.20 ($p = 0.45$) for B-LFW and 0.53 ($p = 0.03$) for beautified Celeb-DF-B genuine frames: which detector accuses most depends on the threshold rule.

S9 Smoothed genuine faces and smoothed forgeries, every form

Table S12: Celeb-DF-v2 test, every fourth frame (4,016 frames: 1,348 genuine, 2,668 forged, 513 videos; a face was found on every frame). Whole-crop forms: bilateral filter at three strengths, 5×5 median, 50% Gaussian blend. Face-local: bilateral filter inside the landmark face hull at blend fraction 0.06 and 0.12, background untouched. Threshold at 5% FPR on untreated genuine frames. Cells: false alarms on smoothed genuine frames (%) and, in parentheses, the change of the miss rate on smoothed forgeries (points). Every arm of ours reproduces its earlier smoothed-forgery scores on the five whole-crop forms (largest frame AUC difference 0.0000).

	false alarms on smoothed genuine frames (%), and in parentheses the change of the miss rate on smoothed forgeries (points)						
Model	bil. light	bil. mid	bil. strong	median 5	Gauss. blend	face-local 0.06	face-local 0.12
Xception [2]	68.0 (-65.8)	76.7 (-59.6)	77.0 (-53.4)	32.1 (-35.3)	20.5 (-22.4)	89.2 (-73.9)	99.3 (-77.5)
EfficientNet-B4 [3]	46.9 (-51.0)	79.7 (-67.6)	85.8 (-68.3)	16.9 (-13.5)	19.1 (-21.2)	64.3 (-63.6)	95.0 (-75.6)
F3Net [4]	74.1 (-76.2)	98.5 (-81.9)	98.7 (-81.9)	31.8 (-40.5)	26.3 (-38.5)	86.3 (-79.5)	99.6 (-82.0)
SPSL [5]	37.2 (-34.6)	63.1 (-49.4)	60.8 (-43.2)	13.7 (-9.4)	7.7 (-1.5)	70.3 (-64.2)	92.1 (-75.7)
UCF [6]	63.4 (-58.5)	24.3 (+3.3)	12.8 (+16.9)	23.1 (-23.3)	19.4 (-21.6)	89.3 (-72.2)	98.4 (-73.7)
RECCE [7]	52.7 (-57.2)	91.8 (-81.2)	94.1 (-83.1)	18.0 (-19.6)	14.6 (-16.1)	80.7 (-80.5)	97.8 (-84.1)
CORE [8]	78.0 (-76.2)	99.3 (-82.3)	99.9 (-82.3)	21.7 (-22.2)	18.9 (-20.0)	85.0 (-79.4)	99.7 (-82.3)
SRM [9]	74.1 (-66.4)	74.0 (-57.9)	63.4 (-47.9)	35.5 (-35.9)	22.4 (-22.8)	81.0 (-76.7)	91.8 (-78.0)
SBI [19]	27.3 (-36.8)	72.3 (-51.6)	91.0 (-59.3)	15.7 (-20.2)	34.8 (-42.1)	96.7 (-61.8)	100.0 (-62.3)
UnivFD‡ [20]	7.9 (+2.9)	40.8 (-27.3)	71.4 (-57.0)	8.2 (+2.8)	3.0 (+8.4)	13.9 (-17.6)	55.2 (-54.6)
NPR‡ [21]	100.0 (-93.6)	100.0 (-93.6)	100.0 (-93.6)	91.8 (-87.3)	98.7 (-92.8)	68.0 (-68.0)	77.7 (-74.9)
Effort [10]	5.8 (+12.0)	27.1 (-9.1)	54.5 (-27.4)	11.3 (-2.0)	9.3 (-9.4)	87.8 (-40.9)	99.9 (-41.3)
Forensics Adapter [11]	27.7 (-16.4)	59.9 (-29.6)	75.3 (-35.3)	18.8 (-8.2)	10.5 (-5.2)	95.8 (-42.8)	100.0 (-42.9)
GenD [12]	32.6 (-24.9)	81.7 (-38.4)	93.7 (-39.5)	28.7 (-20.0)	18.4 (-14.8)	96.1 (-39.9)	100.0 (-39.9)
DFD-HR [13]	21.7 (-13.4)	79.8 (-41.3)	96.5 (-45.5)	17.8 (-7.8)	19.2 (-16.7)	99.3 (-45.7)	100.0 (-45.7)
CADDM [22]	10.8 (+16.9)	61.4 (-32.9)	82.6 (-48.2)	4.5 (+26.2)	6.4 (+0.1)	88.5 (-54.8)	99.9 (-57.3)
LAA-Net [23]	48.5 (-45.4)	92.1 (-54.5)	98.2 (-55.5)	20.1 (-15.7)	25.0 (-32.8)	100.0 (-56.1)	100.0 (-56.1)
P0	0.8 (+27.0)	16.8 (+5.1)	39.6 (-16.0)	1.5 (+19.2)	5.5 (+3.0)	95.8 (-42.1)	100.0 (-42.2)
seed 2	1.6 (+26.7)	18.2 (+6.1)	42.9 (-11.8)	1.9 (+18.0)	4.4 (+8.1)	95.8 (-37.7)	100.0 (-37.7)
seed 3	2.3 (+24.0)	11.3 (+14.3)	25.2 (+4.3)	3.0 (+12.6)	4.9 (+5.6)	96.3 (-36.6)	100.0 (-36.7)
P1-smooth	0.0 (+56.4)	0.0 (+56.4)	0.0 (+56.4)	0.4 (+40.0)	4.0 (+10.8)	0.4 (+41.5)	0.0 (+55.1)
P2	3.0 (+16.5)	31.8 (-17.4)	49.1 (-26.5)	2.5 (+13.2)	3.3 (+7.0)	38.6 (-35.3)	74.5 (-43.0)
seed 2	2.7 (+17.0)	21.1 (-3.4)	36.1 (-13.3)	3.1 (+13.8)	4.5 (+7.3)	34.4 (-28.4)	77.6 (-39.3)
P2-sym	2.4 (+14.8)	5.0 (+14.5)	6.1 (+14.6)	4.5 (+7.6)	3.6 (+9.1)	54.1 (-39.4)	98.7 (-46.2)
seed 2	2.2 (+19.0)	4.2 (+20.6)	5.7 (+17.5)	4.5 (+9.0)	3.0 (+12.7)	37.8 (-31.2)	91.5 (-41.0)
P3	59.6 (-34.6)	99.5 (-38.4)	100.0 (-38.5)	18.8 (-9.3)	6.2 (+3.2)	98.1 (-38.5)	100.0 (-38.5)
OURS, forgery head only†	9.9 (+0.7)	35.5 (-15.4)	45.6 (-15.0)	4.7 (+11.9)	5.4 (+6.2)	51.6 (-32.0)	84.5 (-34.0)
OURS, full model†	34.9 (-25.1)	79.8 (-42.3)	90.1 (-44.3)	8.1 (+4.5)	5.2 (+5.4)	5.8 (+17.6)	44.5 (-14.1)
OURS, full model, sym. view†	1.0 (+28.1)	1.6 (+33.8)	1.7 (+34.1)	2.4 (+12.8)	5.3 (+3.8)	37.9 (-24.9)	80.2 (-36.9)

A detector *accuses* a form if its false alarms on smoothed genuine frames exceed 5% with the interval above it, and *lets through* if its miss rate on smoothed forgeries rises with the interval above zero. Published detectors, by form: light bilateral 14 accuse, 2 both, 1 lets through (Effort); mid bilateral 17 accuse; strong bilateral 16 accuse and 1 both (UCF); median 15 accuse, 1 both, 1 lets through (CADDM); Gaussian blend 16 accuse, 1 lets through (UnivFD); both face-local strengths: 17 accuse. NPR is saturated on whole-crop smoothing (100% false alarms). With both classes smoothed, the published detectors keep a frame AUC of 0.42–0.89 depending on the form, falling toward 0.5 under strong bilateral smoothing for most CNNs; P2, P2-sym and OURS, forgery head only keep 0.75–0.89 except under the stronger face-local form (0.55–0.78).

S10 Texture loss and accusation, every form

Moved from the main text (Sec. III-D) for space; the analysis and its numbers are unchanged.

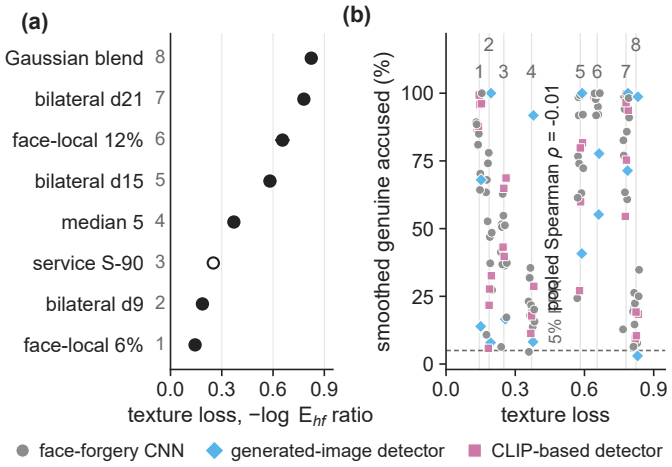


Figure S1: The cue is the smoothing signature rather than blur; Sec. S20 shows that it is a face smoother than its context, not the boundary line. (a) Texture loss of eight smoothing forms, measured as the high-frequency energy removed inside the face hull of 400 genuine Celeb-DF-v2 frames (commercial Smoothing-90 on 200 FFHQ pairs, hollow). (b) False-alarm rate of each published detector at each form against that loss. A Gaussian blend removes the most energy and is accused least; a face-local bilateral filter removes less and is accused by nearly every detector.

S11 Dose response of the CLIP detectors

Table S13: Released Effort, Forensics Adapter, GenD and DFD-HR on the 1,348 genuine frames of Sec. S9. Cells: median shift of the forgery logit (edited minus untreated) with a 95% video-cluster interval (1,000 draws) / the shift in units of the detector's gap between mean forged and mean genuine logit (in parentheses after the name). Vignette: radial darkening at strength 0.1 to 0.8; face-local smoothing at blend fraction 0.03 to 0.12; whole-crop bilateral at three strengths (light and strong shown). Dose response (vignette / face-local / bilateral): the declared rule, a shift toward forged that grows with strength.

Detector (gap)	vignette 0.1	vignette 0.8	face-local 0.03	face-local 0.12	bil. light	bil. strong	dose response
Effort (2.30)	-0.077 [-0.091, -0.064] / -0.03	-0.205 [-0.254, -0.160] / -0.09	+2.396 [+2.202, +2.561] / 1.04	+4.327 [+4.175, +4.433] / 1.88	+0.686 [+0.604, +0.737] / 0.30	+2.280 [+2.125, +2.413] / 0.99	no / yes / yes
Forensics Adapter (3.82)	-0.075 [-0.089, -0.057] / -0.02	-0.359 [-0.428, -0.302] / -0.09	+4.585 [+4.331, +4.874] / 1.20	+9.786 [+9.519, +10.002] / 2.56	+1.798 [+1.645, +1.977] / 0.47	+4.734 [+4.462, +5.047] / 1.24	no / yes / yes
GenD (2.89)	-0.074 [-0.086, -0.061] / -0.03	-0.360 [-0.442, -0.281] / -0.12	+3.382 [+3.205, +3.596] / 1.17	+5.528 [+5.286, +5.700] / 1.91	+1.816 [+1.627, +1.991] / 0.63	+4.489 [+4.312, +4.660] / 1.55	no / yes / yes
DFD-HR (2.05)	-0.001 [-0.010, +0.006] / -0.00	+0.069 [+0.043, +0.095] / 0.03	+3.961 [+3.805, +4.075] / 1.93	+6.060 [+5.954, +6.182] / 2.96	+0.900 [+0.845, +0.965] / 0.44	+3.448 [+3.349, +3.571] / 1.68	yes / yes / yes

Under face-local smoothing the logit rises in at least 97.9% of frames at every level for all four detectors; at the weakest level 61–89% of genuine frames are already above the 5%-FPR threshold, at the strongest 99.9–100%. Under the vignette 18–60% of frames rise; Effort, Forensics Adapter and GenD move toward genuine, and DFD-HR passes the rule with a top-level median of $+0.07$ (3% of its gap). The mean $|\cos|$ between the feature shift and the head direction is 0.04–0.39, except for Forensics Adapter's 192-dimensional head input (0.42–0.69).

S12 Identity-cluster bootstrap

Units: Celeb-DF-B 464 videos in 58 identities; Celeb-DF-v2 filtered 516 videos in 124 units (59 celebrities and 65 YouTube-real singletons); DFD 332 videos in 28 actors; DF40 FF++ domain 140 identities (the Celeb-DF domain has no identity labels and keeps video units). The statistics, draws (1,000, rng 0) and thresholds are those of the paper; only the resampling unit changes. The set tested is: on Celeb-DF-B the beautification-specific miss rise and FA_b (against 0 and 5%) of every detector and the paired AUC_b differences of our full-data models and of GenD and DFD-HR; on the four filter sources the miss and false-alarm rise of every detector and the paired AUC_f differences of our models; on DF40 the paired video-AUC differences of our full-data models. Of these 563 intervals, 24 change category (containing the null or excluding it; Table S14). Identity intervals are 0.86 (Celeb-DF-B, narrower because a person's genuine and forged videos move together), 1.06–1.08 (Celeb-DF-v2), 1.00 (DFD) and 1.30 (DF40, up to 2.8) times as wide as the video intervals (median ratios). An earlier run without the second seeds, which also counted the raw miss rise on Celeb-DF-B

and a merged-identity variant, gave 20 of 456. No paired AUC_b difference of our models against the four CLIP detectors changes category: every identity interval of the full-data models has its lower bound at or above 0.008, and those of P0 at or above 0.006 (Table S7, which uses the same identity draws).

Table S14: The 24 of 563 intervals that change category when identities replace videos as the resampling unit. d_miss, d_fa: rise under the filters; AUC_f: frame AUC under the filters; DF40: video AUC, mean over face-swap (FS) or reenactment (FR) methods, FF++ domain.

Source	quantity	estimate	video CI	identity CI	change
Celeb-DF-B	caddm: d_miss_beauty_only	-2.87	[-5.770, -0.023]	[-5.924, +0.005]	below → contains
Celeb-DF-B	AUC_b gend - fadapter	-0.0208	[-0.045, +0.001]	[-0.041, -0.001]	contains → below
Celeb-DF-v2 + filters (crop)	srm: d_miss	+1.51	[+0.064, +3.269]	[-0.523, +3.848]	above → contains
Celeb-DF-v2 + filters (crop)	FF++-only three-way: d_miss	+3.24	[+0.403, +6.177]	[-0.193, +6.264]	above → contains
Celeb-DF-v2 + filters (crop)	AUC_f OURS, full model - gend	+0.0316	[+0.003, +0.060]	[-0.001, +0.063]	above → contains
Celeb-DF-v2 + filters (full frame)	effnb4: d_miss	+2.47	[+0.118, +4.921]	[-0.120, +5.433]	above → contains
Celeb-DF-v2 + filters (full frame)	AUC_f OURS, forgery head only - gend	+0.0273	[+0.003, +0.051]	[-0.002, +0.054]	above → contains
Celeb-DF-v2 + filters (full frame)	AUC_f sym. view - gend	+0.0249	[+0.003, +0.046]	[-0.001, +0.049]	above → contains
Celeb-DF-v2 + filters (full frame)	AUC_f FF++-only three-way - dfdhr	+0.0274	[+0.004, +0.052]	[-0.002, +0.057]	above → contains
DFD + filters (crop)	AUC_f OURS, full model - fadapter	+0.0197	[+0.000, +0.042]	[-0.004, +0.041]	above → contains
DFD + filters (crop)	AUC_f FF++-only three-way - fadapter	+0.0177	[+0.000, +0.037]	[-0.002, +0.036]	above → contains
DFD + filters (full frame)	xception: d_miss	+1.96	[-0.688, +5.277]	[+0.190, +4.295]	contains → above
DFD + filters (full frame)	AUC_f sym. view s2 - gend	+0.0145	[-0.001, +0.030]	[+0.000, +0.028]	contains → above
DF40	OURS, full model - gend: ff/FS	-0.0149	[-0.023, -0.007]	[-0.030, +0.001]	below → contains
DF40	OURS, full model s2 - effort: ff/FS	+0.0154	[+0.006, +0.025]	[-0.006, +0.036]	above → contains
DF40	OURS, full model s2 - effort: ff/FR	-0.0280	[-0.045, -0.010]	[-0.070, +0.011]	below → contains
DF40	OURS, full model s2 - fadapter: ff/FS	+0.0117	[+0.003, +0.021]	[-0.005, +0.029]	above → contains
DF40	OURS, forgery head only - effort: ff/FR	+0.0343	[+0.017, +0.051]	[-0.005, +0.076]	above → contains
DF40	sym. view s2 - effort: ff/FS	+0.0137	[+0.004, +0.024]	[-0.007, +0.035]	above → contains
DF40	sym. view s2 - fadapter: ff/FS	+0.0099	[+0.002, +0.019]	[-0.005, +0.027]	above → contains
DF40	sym. view s2 - gend: ff/FS	-0.0086	[-0.016, -0.001]	[-0.023, +0.005]	below → contains
DF40	OURS, forgery head only s2 - effort: ff/FS	+0.0172	[+0.009, +0.027]	[-0.001, +0.035]	above → contains
DF40	OURS, forgery head only s2 - fadapter: ff/FS	+0.0135	[+0.005, +0.022]	[-0.001, +0.029]	above → contains
DF40	OURS, forgery head only s2 - fadapter: ff/FR	+0.0281	[+0.014, +0.041]	[-0.006, +0.064]	above → contains

S13 The decomposition of a commercial render

Algorithm S1: dense optical flow on locally normalised luminance gives the warp f between an original O and its render R, the photometric residual is taken in the geometry of O, and four single-operation components are formed, each with its measured strength and exact footprint. Applying the four in sequence reproduces R at 108 dB PSNR.

Algorithm S1 Single-operation components of a commercial render

Require: original O, render R, eye mask E, σ, noise floors τ_k

$f \leftarrow \mathrm{DIS}\big(\mathrm{LCN}(O), \mathrm{LCN}(R)\big)$

$P \leftarrow R \circ (\mathrm{id} - f^{-1}) - O$

$C_{\text{eye}} \leftarrow \mathcal{W}(O; E \odot f)$; $\quad C_{\text{contour}} \leftarrow \mathcal{W}(O; (1-E) \odot f)$

$C_{\text{tone}} \leftarrow O + G_\sigma * P$; $\quad C_{\text{texture}} \leftarrow O + P - G_\sigma * P$

for $k \in \{\text{eye}, \text{contour}, \text{tone}, \text{texture}\}$ **do**

  $q_k \leftarrow \mathrm{Measure}_k(C_k, O)$; **if** $q_k \geq \tau_k$ **keep** $(C_k,\ e_k,\ M(C_k))$

end for

The loss is $\mathcal{L} = \mathrm{BCE}(z_f, y_f) + \mathrm{BCE}(z_b, y_b) + \mathbf{1}[y_b]\,\mathrm{BCE}(\hat{s}, s) + \mathbf{1}[M \text{ exact}]\,\big(\mathrm{BCE}(\hat{m}, M) + \mathrm{Dice}(\hat{m}, M)\big)$, with $\hat{s}$ the operation-presence logits.

S14 Exposure arms on every source

Table S15: Which class receives beautified training views. FF++ only, forged head only, everything else identical; three seeds. Celeb-DF-B under the one rule of the paper.

Arm	Beautified training views	AUC_b (3 seeds)	$\Delta\text{MISS}_{\text{beauty}}$ (3 seeds)	FA_b
P0	no beautified views	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	5.1 / 5.6 / 6.8
P1	beautified genuine faces only	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	4.5 / 3.9 / 3.7
P2	beautified genuine and forged faces	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	5.6 / 5.3 / 5.4
Paired differences, 95% video-cluster intervals (seeds 1 / 2 / 3)				
P1 − P0	$\Delta\text{MISS}_{\text{beauty}}$	+4.5 [1.5, 7.6] / +7.5 [4.4, 10.8] / +9.5 [6.7, 12.3]		
P2 − P0	$\Delta\text{MISS}_{\text{beauty}}$	-7.2 [-9.4, -5.0] / -4.0 [-6.1, -2.0] / +2.0 [0.1, 4.0]		
P2 − P1	$\Delta\text{MISS}_{\text{beauty}}$	-11.7 [-14.5, -8.8] / -11.5 [-14.5, -8.9] / -7.5 [-10.0, -5.0]		

Table S16: P0 none, P1 genuine only, P2 both; FF++ only, forged head only; values per seed. Effort scheme: the SVD-residual tuning of the released Effort code in place of LoRA; P1 was not trained under it. Frame AUC; ΔMISS at the 5%-FPR threshold of untreated genuine frames.

		Celeb-DF-B			Celeb-DF-v2 filters		DFD filters	
Tuning	Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	AUC_f	ΔMISS	AUC_f	ΔMISS
LoRA	P0	0.927 / 0.937 / 0.944	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	0.852 / 0.858 / 0.855	+7.1 / +4.4 / +8.3	0.924 / 0.928 / 0.932	+9.3 / +7.2 / +8.7
LoRA	P1	0.928 / 0.931 / 0.939	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	0.798 / 0.820 / 0.838	+16.6 / +17.7 / +16.3	0.905 / 0.911 / 0.913	+11.2 / +9.3 / +9.2
LoRA	P2	0.929 / 0.933 / 0.935	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	0.848 / 0.850 / 0.856	+0.3 / +3.0 / -0.9	0.929 / 0.929 / 0.927	+2.5 / +1.8 / -0.5
Effort scheme	P0	0.943 / 0.948	0.882 / 0.886	+10.1 / +10.7	0.876 / 0.886	+2.3 / -1.3	0.942 / 0.947	+8.7 / +7.4
Effort scheme	P2	0.938 / 0.943	0.889 / 0.890	+5.5 / +5.5	0.872 / 0.879	-7.6 / -8.4	0.948 / 0.953	+3.0 / +2.4

Bootstrap p-values for the nine exposure contrasts on Celeb-DF-B (P1−P0, P2−P0 and P2−P1 on $\Delta\text{MISS}_{\text{beauty}}$, three seeds, 1,000 identical video draws, $p = 2\min(\Pr[d \le 0], \Pr[d \ge 0])$ floored at 10^{-3}), Holm-adjusted over the nine: P1−P0 is $+4.5$, $+7.5$ and $+9.5$ points (adjusted $p = 0.009$ each); P2−P1 is -11.7, -11.5 and -7.5 (0.009 each); P2−P0 is -7.2 and -4.0 (0.009) and $+2.0$ on the third seed (0.044). The family-wise picture is the one the paper states: P1 is worse than P0 and P2 in every seed, and P2 against P0 changes sign across seeds. Paired video-cluster intervals for P2 minus P0 on the two filter sources: the miss rise falls by 5.0 to 9.9 points in nine of ten seed-by-source cells (intervals below zero; the tenth is -1.4 $[-3.3, 0.6]$), the false-alarm rise grows by 2.0 to 4.6 points in the same nine, and the frame AUC under the filters changes by -0.008 to $+0.006$ (intervals contain zero except for the Effort scheme on DFD, $+0.006$). For P1 minus P0 on Celeb-DF-v2 the miss rise grows by 13.3 and 8.0 points and the AUC under the filters falls by 0.038 and 0.017.

S15 The forged-only arm

Table S17: The four exposure arms, seed 20260929: Smoothing-90 accused at the 5%-FPR threshold of the originals with a 95% interval over originals, share of renders above their own original, false alarms on whole-crop- and skin-only-smoothed genuine Celeb-DF-v2 frames (%), and DF40 reenactment video AUC (FF++ / Celeb-DF domain). P1 was not scored on the skin-only forms (P1-smooth: 0.4 and 0.0%).

Arm	Smoothing-90 accused (%) [CI]	above own (%)	FA bil. light/mid/strong/med./blend	face-local 0.06/0.12	DF40 FR ff / cdf
P0	54.2 [50.5, 58.3]	98.5	0.8 / 16.8 / 39.6 / 1.5 / 5.5	95.8 / 100.0	0.906 / 0.692
P1	3.7 [2.9, 4.4]	43.5	0.0 / 0.0 / 0.0 / 0.7 / 4.7	–	0.807 / 0.569
P2	11.7 [9.7, 14.3]	84.1	3.0 / 31.8 / 49.1 / 2.5 / 3.3	38.6 / 74.5	0.809 / 0.644
P3	63.8 [59.0, 67.8]	99.2	59.6 / 99.5 / 100.0 / 18.8 / 6.2	98.1 / 100.0	0.870 / 0.753

Declared reading: (i) the sign rule is complete if P3 minus P0 on Smoothing-90 is not wholly below -5 and P3's false alarms on mid and strong whole-crop-smoothed genuine frames are not lower than P0's; (ii) forged-only views also remove the accusation if P3 accuses at most 15%. P3 minus P0 is $+9.6$ $[+6.0, +13.4]$ (P3 minus P2 $+52.1$ $[+47.5, +55.8]$) and the false alarms are 99.5 and 100% against 16.8 and 39.6%: (i) holds, (ii) does not. P3 mirrors P1: forgeries under the filters are missed less often than untreated ones (-25.1 points on Celeb-DF-v2, -6.6 on DFD, against $+4.4$ and $+8.5$ for P0),

the miss rate on beautified Celeb-DF-B forgeries rises by 1.0 point (P0 +23.3), and beautified Celeb-DF-B genuine frames are accused at 24.6% (P0 5.1%). Untreated detection is not lower: Celeb-DF-v2 video AUC 0.943 (P0 0.938), DFD 0.982 (0.981), DF40 face swap 0.970 and 0.937. On the service, its native decision sends 93.9% of the Smoothing-90 renders to forged. One seed.

S16 The colour-filter probe, per filter

The probed model is the FF++-only three-way CLIP model ("FF++-only three-way" in Tables S5 and S6), trained with random colour grading and 14 colour filters as beautified views; the training-time subspace below exists because those views were in its training. No published detector was probed with this subspace analysis. The dose response of the released CLIP detectors (Sec. S11) shows that a vignette in isolation does not move three of them toward forged, so the radial-overlay effect below is a property of this model, not a general account of the filter result.

Table S18: FF++ validation frames under the eleven held-out filters: cosine of the mean feature shift with the forged weight vector w, mean change of the forged logit on genuine and forged frames, and share of the shift energy inside the 16-dimensional subspace spanned by training-time beautifications.

Held-out filter	n	$\cos(w, \bar{\Delta})$	Δlogit genuine	Δlogit forged	energy in V_{16}
earlybird	88	0.20	+7.84	+1.10	0.86
mayfair	94	0.33	+5.64	+1.80	0.83
kelvin	92	0.04	+2.40	-0.71	0.90
clarendon	99	0.07	+1.90	+0.40	0.86
lofi	95	0.06	+1.86	+0.17	0.87
moon	89	0.06	+1.75	-0.10	0.75
perpetua	107	0.09	+0.99	+0.21	0.77
inkwell	95	0.09	+0.86	-0.25	0.63
reyes	104	0.02	+0.76	-0.27	0.89
slumber	101	0.00	+0.43	-0.28	0.87
brooklyn	95	0.00	+0.14	-0.04	0.88

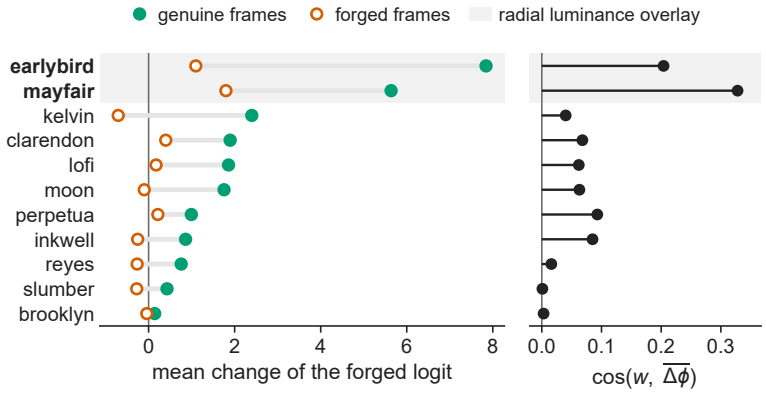


Figure S2: Mean change of the forged logit under each held-out filter on genuine (filled) and forged (open) frames (left), and cosine of the mean feature shift with w (right). Shaded rows: the two filters with a radial luminance overlay.

Table S19: Linear erasure. The feature shifts caused by training-time beautifications are decomposed by SVD on training frames, the first k directions are projected out, and a logistic forged head is refitted on training features. FA and MISS at the 5%-FPR threshold of clean genuine frames, pooled over the eleven held-out filters. The control is the refit without erasure ($k = 0$), not the trained head.

Forged head	AUC clean	AUC filtered	FA filtered	MISS filtered
trained head	0.987	0.953	21.5	5.9
refit, 0 directions erased	0.985	0.916	15.8	16.6
refit, 2 directions erased	0.933	0.907	18.5	15.1
refit, 8 directions erased	0.932	0.844	18.3	27.6
refit, 32 directions erased	0.919	0.804	19.2	32.1

The two leading directions hold 79% of the shift energy of training-time beautifications. Against the refit without erasure, removing them costs 0.052 clean AUC and raises the pooled false alarms under the filters from 15.8 to 18.5% (18.3 and 19.2% for 8 and 32 directions); per filter the false-alarm rate after erasure ranges from 17.5 to 27.5%. The directions along which beautification moves the feature carry forgery evidence for this model.

S17 Skin smoothing: the chain of pre-registered arms

All arms are FF++-only models with the forged head only; they share the architecture, data, steps and checkpoint rule of the exposure arms and differ only in which class receives beautified views and what those views contain. No image of any retouching service is used. The scripted smoothing is a bilateral filter blended inside a feathered face hull, the same kind of edit as the service's (face-local and edge-preserving: high-frequency energy on flat skin 0.68 at the top of the training range against 0.60 for the service's level 90, background 1.00 against 0.96). Its strength range was set to cover the levels measured on the two sister services in RetouchingFFHQ while the test service's measurements were on disk.

Table S20: Commercial skin smoothing, every detector and arm: 2,210 FFHQ originals and their renders. Smoothing-90: share of renders scored above their own original (50 means no shift), medians of $P(\text{fake})$, and share above the threshold that accuses 5% of the originals. All renders: the four operations at three levels (26,519 images). NPR scores originals and renders at 1.0.

	Smoothing-90 renders			all renders	
Detector	above own original (%)	median $P(\text{fake})$ orig. $\rightarrow$ render	above 5%-FPR thr. (%)	above 5%-FPR thr. (%)	AUROC render vs orig.
Xception [2]	95.2	0.291 $\rightarrow$ 0.944	51.6	12.2	0.563
EfficientNet-B4 [3]	93.7	0.316 $\rightarrow$ 0.964	36.3	9.4	0.545
F3Net [4]	97.8	0.131 $\rightarrow$ 0.974	50.6	11.8	0.568
SPSL [5]	97.3	0.463 $\rightarrow$ 0.963	51.2	12.1	0.567
UCF [6]	95.4	0.031 $\rightarrow$ 0.984	41.4	11.2	0.575
RECCE [7]	96.5	0.120 $\rightarrow$ 0.944	36.8	9.5	0.558
CORE [8]	96.6	0.249 $\rightarrow$ 0.947	54.8	12.4	0.569
SRM [9]	99.3	0.461 $\rightarrow$ 0.668	62.8	13.9	0.570
SBI [19]	96.2	0.208 $\rightarrow$ 0.991	37.3	11.3	0.582
UnivFD‡ [20]	90.4	0.005 $\rightarrow$ 0.019	16.6	7.3	0.526
NPR‡ [21]	n/a (saturated)	n/a	n/a	n/a	n/a
Effort [10]	95.6	0.319 $\rightarrow$ 0.595	43.2	14.8	0.599
Forensics Adapter [11]	93.3	0.058 $\rightarrow$ 0.354	39.7	10.2	0.541
GenD [12]	98.5	0.228 $\rightarrow$ 0.921	68.7	16.8	0.596
DFD-HR [13]	99.0	0.210 $\rightarrow$ 0.711	64.9	15.7	0.554
CADDM [22]	60.9	0.538 $\rightarrow$ 0.745	6.3	5.6	0.523
LAA-Net [23]	89.2	0.234 $\rightarrow$ 0.969	17.2	7.6	0.578
Ours, forgery head only	70.5	0.000 $\rightarrow$ 0.001	9.1	5.2	0.490
Ours, full model	32.5	0.002 $\rightarrow$ 0.001	8.6	4.7	0.371
Ours, full model, sym. view	30.0	0.003 $\rightarrow$ 0.001	8.1	4.8	0.375
FF++-only exposure arms (seed 1)					
P0	98.5	0.049 $\rightarrow$ 0.968	54.2	13.5	0.576
P2	84.1	0.009 $\rightarrow$ 0.116	11.7	5.9	0.528
P2 no smoothing	98.6	0.017 $\rightarrow$ 0.988	55.3	12.7	0.553
P2 smoothing only	27.3	0.053 $\rightarrow$ 0.020	2.3	3.7	0.449
P1	43.5	0.004 $\rightarrow$ 0.004	3.7	5.3	0.500
P1 smoothing only	2.6	0.074 $\rightarrow$ 0.000	0.7	4.9	0.437
P2 symmetric smoothing	53.7	0.017 $\rightarrow$ 0.022	5.9	4.9	0.488
FF++ only, Effort's tuning scheme (seed 1)					
Effort scheme, no beautified views	99.1	0.028 $\rightarrow$ 0.963	56.9	13.3	0.587
Effort scheme, symmetric smoothing	88.4	0.029 $\rightarrow$ 0.095	14.5	7.2	0.541
released Effort fine-tuned, no beautified views	98.5	0.091 $\rightarrow$ 0.306	55.3	14.4	0.564
released Effort fine-tuned, symmetric smoothing	96.6	0.080 $\rightarrow$ 0.174	39.9	11.5	0.553

Table S21: Paired shift of the forgery logit, Smoothing-90 render minus its own original (2,210 pairs): counts of pairs moving up and down, median shift with a 95% bootstrap interval over originals (1,000 draws), and the two-sided sign test with ties dropped. Probabilities are clipped to $[10^{-7}, 1-10^{-7}]$ before the logit. NPR is saturated and omitted.

Model	up / down / tie	median Δlogit [95% CI]	sign test p
Xception [2]	2105 / 105 / 0	+3.32 [+3.17, +3.45]	$< 10^{-10}$
EfficientNet-B4 [3]	2070 / 140 / 0	+3.48 [+3.33, +3.57]	$< 10^{-10}$
F3Net [4]	2161 / 49 / 0	+4.92 [+4.74, +5.08]	$< 10^{-10}$
SPSL [5]	2151 / 59 / 0	+3.05 [+2.95, +3.12]	$< 10^{-10}$
UCF [6]	2108 / 102 / 0	+5.47 [+5.20, +5.73]	$< 10^{-10}$
RECCE [7]	2133 / 77 / 0	+3.82 [+3.70, +3.96]	$< 10^{-10}$
CORE [8]	2135 / 75 / 0	+3.90 [+3.74, +4.03]	$< 10^{-10}$
SRM [9]	2194 / 16 / 0	+0.68 [+0.65, +0.71]	$< 10^{-10}$
SBI [19]	2127 / 83 / 0	+4.88 [+4.72, +5.04]	$< 10^{-10}$
UnivFD‡ [20]	1998 / 212 / 0	+1.22 [+1.17, +1.28]	$< 10^{-10}$
Effort [10]	2113 / 97 / 0	+0.96 [+0.90, +1.02]	$< 10^{-10}$
Forensics Adapter [11]	2063 / 147 / 0	+1.72 [+1.59, +1.87]	$< 10^{-10}$
GenD [12]	2176 / 34 / 0	+3.27 [+3.19, +3.35]	$< 10^{-10}$
DFD-HR [13]	2189 / 21 / 0	+2.06 [+1.93, +2.15]	$< 10^{-10}$
CADDM [22]	1345 / 865 / 0	+0.67 [+0.53, +0.81]	$< 10^{-10}$
LAA-Net [23]	1971 / 239 / 0	+3.60 [+3.36, +3.79]	$< 10^{-10}$
OURS, forgery head only	1557 / 642 / 11	+0.41 [+0.38, +0.44]	$< 10^{-10}$
second seed	1508 / 694 / 8	+0.19 [+0.16, +0.19]	$< 10^{-10}$
OURS, full model	718 / 1490 / 2	-1.20 [-1.28, -1.12]	$< 10^{-10}$
second seed	913 / 1296 / 1	-0.53 [-0.60, -0.41]	$< 10^{-10}$
OURS, full model, sym. view	662 / 1545 / 3	-1.09 [-1.19, -1.00]	$< 10^{-10}$
P0	2176 / 34 / 0	+5.17 [+5.05, +5.38]	$< 10^{-10}$
P2	1859 / 347 / 4	+1.81 [+1.68, +1.97]	$< 10^{-10}$
P1	962 / 1245 / 3	-0.22 [-0.28, -0.16]	1.8×10^{-9}
P1-smooth	57 / 2153 / 0	-5.00 [-5.16, -4.88]	$< 10^{-10}$
P2-smooth	604 / 1602 / 4	-0.91 [-0.97, -0.84]	$< 10^{-10}$
P2-sym	1187 / 1017 / 6	+0.09 [+0.03, +0.12]	3.2×10^{-4}
P2-sym, seed 2	1144 / 1059 / 7	+0.03 [-0.00, +0.09]	0.07

The published detectors and the no-exposure arm move the logit of a smoothed genuine face up by 0.7 to 5.5. OURS, full model, its variant and the genuine-only or form-asymmetric arms move it down, which is the evasion direction; OURS, forgery head only moves it up by 0.41 and 0.19 (two seeds). Only P2-sym is close to no shift, with a small upward median on its first seed ($p = 3.2 \times 10^{-4}$) and none detectable on its second ($p = 0.07$). A 25–75% band on the share of pairs, the bar used for P2-sym and the variant, is wide: it admits OURS, full model's 32.5%, which the sign test shows is a shift.

The no-exposure arm reproduces the published detectors. P0 accuses 54.2, 56.1 and 43.5% (three seeds) of the Smoothing-90 renders at the 5%-FPR threshold of the service's originals, scores the render above its own original in 97.0–99.0% of pairs, and moves the median $P(\text{fake})$ from 0.05 to 0.97 (seed 1). P0 is at the 5% floor on the other nine operation-by-level cells of the service (whitening 4.5–7.5%, face slimming 4.5–6.1%, eye enlarging 4.8–5.9%) and dose-dependent on smoothing (12.6–17.0, 29.5–40.8 and 43.5–56.1% at levels 30, 60 and 90). P0 trains with label-symmetric Gaussian blur, down-up scaling and JPEG on half of its triplets, so plain blur or re-compression in training does not prevent the accusation. With beautified views on genuine faces only (P1) the Smoothing-90 rate falls to 3.7, 7.1 and 4.6%; with views on both classes (P2) to 11.7, 7.1 and 6.9% (paired differences against P0 of -37 to -51 points, every interval below zero). The rate holds under every threshold rule tried for these arms (1% and 10% FPR, a fixed 0.5, thresholds set on Celeb-DF-v2 or DFD genuine frames). P2 still scores the render above its original in 72–84% of pairs (AUROC 0.63–0.70). Both arms pay for it on untreated Celeb-DF-v2: frame AUC -0.012 to -0.021 for P2 and -0.025 to -0.030 for P1 against P0 on every seed, and 5–8 points of miss rate at 5% FPR. This first analysis was post hoc, and its per-epoch curve for the both-class arm (27% at epoch 2, 9–12% from epoch 5, with checkpoints chosen on FF++ validation AUC by margins of 0.0003–0.002) is disclosed for that reason. Inside Effort's SVD-residual scheme the P2 views leave 30.0 and 38.9% (P0 under that scheme 56.9 and 63.2%), and two epochs of fine-tuning the released Effort checkpoint with them leave 52.7% against 55.3% without them.

No other view substitutes for smoothing. Declared reading: if the arm stays at 30% or more on both seeds, the effect is exposure to smoothing itself. P2-nosmooth removes smoothing from every beautified view of P2 and keeps colour grades, the fourteen colour filters, whitening, eye and jaw edits, and the shared degradation. Smoothing-90 is then accused at 55.3% [51.9, 58.6] and 67.1% (two seeds), against P0 of the same seed $+1.2$ [-2.2, 4.0] and $+11.1$ [7.0, 14.8], and the render is ranked above its original in 98.6 and 99.4% of pairs. The genuine-only version without smoothing gives 55.9%

Table S22: The exposure table of the paper (Table III) with every column; values per seed (1 / 2 / 3 where trained). Arm names in the paper: P2 no smoothing = P2-nosmooth, P2 smoothing only = P2-smooth, P1 smoothing only = P1-smooth, P2 symmetric smoothing = P2-sym. Untreated video AUC; Celeb-DF-B frame AUC; DF40 reenactment video AUC.

	commercial Smoothing-90		smoothed Celeb-DF-v2 (strong bil.)		untreated video AUC		Celeb-DF-B (frame)		DF40 FR video AUC	
Arm	accused (%)	above own orig. (%)	miss rise	frame AUC, both smoothed	Celeb-DF-v2	DFD	AUC_b	$\Delta MISS_{beauty}$	FF++ dom.	Celeb-DF dom.
P0	54.2 / 56.1 / 43.5	98.5 / 99.0 / 97.0	-16.0 / -11.8 / +4.3	0.747 / 0.736 / 0.736	0.938 / 0.938 / 0.940	0.981 / 0.979 / 0.978	0.860 / 0.883 / 0.883	+17.0 / +14.2 / +10.8	0.906 / 0.877 / 0.873	0.692 / 0.720 / 0.633
P1	3.7 / 7.1 / 4.6	43.5 / 67.5 / 64.3	+51.2 / – / –	0.587 / – / –	0.900 / 0.910 / 0.923	0.978 / 0.975 / 0.971	0.844 / 0.851 / 0.864	+21.5 / +21.8 / +20.4	0.807 / – / –	0.569 / – / –
P2	11.7 / 7.1 / 6.9	84.1 / 72.4 / 75.5	-26.5 / -13.3 / –	0.740 / 0.746 / –	0.917 / 0.920 / 0.935	0.973 / 0.975 / 0.972	0.879 / 0.893 / 0.877	+9.8 / +10.3 / +12.8	0.809 / 0.785 / –	0.644 / 0.643 / –
P3	63.8	99.2	-38.5	0.743	0.943	0.982	0.857	-2.3	0.870	0.753
P2-nosmooth	55.3 / 67.1	98.6 / 99.4	-28.9 / -33.1	0.732 / 0.695	0.913 / 0.924	0.980 / 0.973	0.879 / 0.873	+12.9 / +12.9	0.858 / –	0.705 / –
P2-smooth	2.3 / 5.3	27.3 / 51.7	-48.2 / -44.6	0.785 / 0.734	0.904 / 0.926	0.967 / 0.966	0.865 / 0.870	+10.2 / +13.7	0.779 / 0.833	0.593 / 0.629
P1-smooth	0.7 / 1.0 / 0.9	2.6 / 6.5 / 5.4	+56.4 / +60.0 / +62.9	0.643 / 0.579 / 0.596	0.922 / 0.929 / 0.939	0.974 / 0.977 / 0.964	0.887 / 0.896 / 0.893	+8.6 / +8.5 / +7.3	0.871 / 0.869 / 0.900	0.673 / 0.689 / 0.679
P2-sym	5.9 / 6.1	53.7 / 51.8	+14.6 / +17.5	0.767 / 0.796	0.919 / 0.934	0.972 / 0.968	0.866 / 0.888	+14.2 / +10.4	0.871 / 0.796	0.670 / 0.646
Effort scheme, P0	56.9 / 63.2	99.1 / 99.1	-32.2 / –	0.719 / –	0.944 / 0.954	0.987 / 0.991	0.882 / 0.886	+10.1 / +10.7	– / –	– / –
Effort scheme, P2-sym	14.5	88.4	+24.9	0.757	0.935	0.986	0.890	+10.6	0.822	0.614
released Effort fine-tuned, no views	55.3	98.5	-39.9	0.754	0.926	0.976	0.805	+5.8	–	–
released Effort fine-tuned, P2-sym view	39.9	96.6	-19.0	0.740	0.921	0.974	0.791	+7.5	0.805	0.705
OURS, forgery head only	9.1 / 6.5	70.5 / 68.2	-15.0 / +2.2	0.754 / 0.810	0.956 / 0.963	0.987 / 0.978	0.896 / 0.904	+7.0 / +5.7	0.842 / 0.856	0.647 / 0.698
OURS, full model	8.6 / 12.5	32.5 / 41.3	-44.3 / -46.7	0.688 / 0.711	0.947 / 0.946	0.978 / 0.981	0.866 / 0.883	+6.7 / +8.3	0.718 / 0.780	0.502 / 0.559
OURS, full model, sym. view	8.1 / 17.8	30.0 / 46.2	+34.1 / +11.6	0.796 / 0.784	0.954 / 0.953	0.983 / 0.987	0.868 / 0.888	+10.0 / +3.1	0.738 / 0.718	0.539 / 0.504

Table S23: Smoothed forgeries and smoothed genuine frames. Celeb-DF-v2 test frames (every fourth of the 32 per video), untreated and under a whole-crop bilateral filter at three strengths, a 5×5 median filter and a 50% blend with a Gaussian blur; the last two forms were shown to no arm. Threshold at 5% FPR on untreated genuine frames. Miss rise: change of the miss rate on forgeries after smoothing, with 95% video-cluster intervals. Last column: frame AUC with both classes under the strong bilateral filter.

		miss rise (points, 95% CI)					false alarms, smoothed genuine (%)					
Arm	miss untr.	bil. light	bil. mid	bil. strong	median 5	Gauss. blend	bil. light	bil. mid	bil. strong	median 5	Gauss. blend	AUC both smoothed (strong)
P0	42.2	+27.0 [+24.4, +29.7]	+5.1 [-0.3, +10.5]	-16.0 [-21.5, -10.2]	+19.2 [+17.1, +21.2]	+3.0 [+1.4, +5.1]	0.8	16.8	39.6	1.5	5.5	0.747
seed 2	37.7	+26.7 [+24.1, +29.9]	+6.1 [+2.2, +11.8]	-11.8 [-15.5, -5.4]	+18.0 [+16.0, +20.4]	+8.1 [+6.6, +9.9]	1.6	18.2	42.9	1.9	4.4	0.736
seed 3	36.7	+24.0 [+21.4, +26.9]	+14.3 [+8.8, +17.9]	+4.3 [-2.4, +11.0]	+12.6 [+10.6, +14.6]	+5.6 [+4.2, +6.9]	2.3	11.3	25.2	3.0	4.9	0.736
P2	46.8	+16.5 [+14.2, +19.4]	-17.4 [-20.7, -13.1]	-26.5 [-29.7, -21.9]	+13.2 [+11.7, +15.8]	+7.0 [+6.0, +8.9]	3.0	31.8	49.1	2.5	3.3	0.740
P2 no smoothing	46.7	+19.8 [+16.8, +22.0]	-12.2 [-16.3, -6.5]	-28.9 [-32.8, -24.9]	+8.1 [+6.4, +9.8]	+2.2 [+0.7, +3.5]	2.3	25.6	54.4	3.9	4.7	0.732
P2 smoothing only	49.5	-34.3 [-37.6, -30.4]	-47.3 [-51.6, -40.2]	-48.2 [-53.1, -40.8]	+17.2 [+14.7, +20.2]	+10.9 [+9.2, +13.6]	47.8	82.3	88.3	1.7	2.4	0.785
P1	48.8	+45.9 [+39.7, +51.1]	+51.1 [+43.6, +57.2]	+51.2 [+43.7, +57.5]	+29.3 [+25.1, +32.4]	+6.4 [+4.8, +7.6]	0.0	0.0	0.0	0.7	4.7	0.587
P1 smoothing only	43.6	+56.4 [+49.9, +63.7]	+56.4 [+50.0, +63.9]	+56.4 [+50.0, +63.9]	+40.0 [+35.8, +44.4]	+10.8 [+9.1, +12.6]	0.0	0.0	0.0	0.4	4.0	0.643
seed 2	40.0	+58.7 [+52.3, +65.1]	+60.0 [+53.7, +66.7]	+60.0 [+53.7, +66.7]	+38.6 [+34.6, +42.6]	+13.9 [+12.3, +16.1]	0.0	0.0	0.0	0.7	3.1	0.579
seed 3	37.1	+59.3 [+53.9, +64.5]	+62.8 [+57.4, +68.6]	+62.9 [+57.5, +68.7]	+38.0 [+33.8, +41.7]	+13.5 [+11.9, +15.3]	0.0	0.0	0.0	0.5	2.7	0.596
P2 symmetric smoothing	46.2	+14.8 [+12.4, +17.0]	+14.5 [+11.4, +16.9]	+14.6 [+11.3, +17.6]	+7.6 [+5.4, +9.3]	+9.1 [+7.7, +10.6]	2.4	5.0	6.1	4.5	3.6	0.767
seed 2	42.0	+19.0 [+16.7, +21.5]	+20.6 [+17.3, +23.6]	+17.5 [+14.7, +20.5]	+9.0 [+7.2, +10.6]	+12.7 [+11.0, +14.3]	2.2	4.2	5.7	4.5	3.0	0.796
Effort scheme, no beautified views	39.7	+19.0 [+16.2, +21.7]	-11.3 [-15.8, -6.3]	-32.2 [-36.2, -27.5]	+11.2 [+8.4, +13.5]	-2.1 [-3.9, +0.1]	3.6	35.4	75.1	4.8	7.6	0.719
Effort scheme, symmetric smoothing	41.7	+22.4 [+20.0, +25.1]	+25.0 [+21.3, +28.5]	+24.9 [+19.3, +28.5]	+5.8 [+3.4, +7.9]	+7.2 [+5.8, +8.7]	2.0	4.2	6.1	7.3	4.3	0.757
released Effort fine-tuned, no beautified views	42.9	+17.2 [+12.3, +20.0]	-21.6 [-26.0, -14.6]	-39.9 [-44.7, -34.7]	+9.8 [+6.3, +12.2]	+1.1 [-3.2, +4.3]	4.2	39.2	81.2	7.7	6.4	0.754
released Effort fine-tuned, symmetric smoothing	43.7	+23.0 [+18.3, +26.4]	+6.6 [-1.3, +13.5]	-19.0 [-26.4, -10.1]	+10.0 [+5.9, +12.6]	-1.2 [-6.6, +2.9]	2.2	15.7	43.0	7.0	7.3	0.740
OURS, forgery head only	36.5	+0.7 [-3.2, +7.5]	-15.4 [-21.1, -2.5]	-15.0 [-20.8, -0.7]	+11.9 [+9.9, +14.0]	+6.2 [+5.1, +8.0]	9.9	35.5	45.6	4.7	5.4	0.754
second seed	35.4	+22.4 [+19.6, +24.5]	+6.3 [+1.7, +10.1]	+2.2 [-3.1, +6.7]	+13.6 [+12.0, +16.0]	+8.4 [+6.6, +9.7]	2.2	9.3	17.2	3.4	3.6	0.810
OURS, full model	47.0	-25.1 [-28.6, -21.8]	-42.3 [-46.3, -36.6]	-44.3 [-49.1, -38.3]	+4.5 [+1.7, +6.9]	+5.4 [+3.7, +6.6]	34.9	79.8	90.1	8.1	5.2	0.688
second seed	47.3	-15.0 [-18.7, -10.7]	-43.9 [-49.2, -36.5]	-46.7 [-54.9, -38.4]	+9.3 [+7.1, +11.7]	+0.1 [-1.0, +2.1]	22.0	84.1	96.1	3.6	5.6	0.711
OURS, full model, sym. view	43.5	+28.1 [+25.2, +30.6]	+33.8 [+29.9, +36.8]	+34.1 [+30.2, +37.1]	+12.8 [+11.0, +15.1]	+3.8 [+2.2, +5.1]	1.0	1.6	1.7	2.4	5.3	0.796
second seed	45.1	+9.7 [+6.9, +12.2]	+13.8 [+9.9, +17.4]	+11.6 [+7.4, +15.9]	-0.1 [-2.5, +2.6]	+2.7 [+1.1, +4.1]	3.8	5.9	8.4	6.3	5.1	0.784

($+1.8$ [-1.8, 5.2] against P0). The declared reading holds within the views we trained. These arms still lose untreated Celeb-DF-v2 video AUC against P0 (0.913 and 0.924 against 0.938 and 0.938), so that cost is not caused by smoothing alone.

Smoothing as the only view, on both classes (P2-smooth). Bars, against P0 of the same seed, on both seeds: (a) Smoothing-90 at most 15% at the 5%-FPR threshold of the originals; (b) untreated Celeb-DF-v2 and DFD video AUC not lower by more than 0.010; (c) Celeb-DF-B frame AUC under the presets not lower by more than 0.010. Smoothing-90 falls to 2.3% [1.7, 2.9] and 5.3% [4.4, 7.0], -51.9 [-56.1, -48.3] and -50.7 [-55.3, -46.5] against P0: (a) is met on both seeds. Bar (b) is not met: untreated Celeb-DF-v2 video AUC is 0.904 and 0.926 against 0.938 and 0.938, untreated DFD 0.967 and 0.966 against 0.981 and 0.979. Bar (c) is 0.865 and 0.870 against 0.860 and 0.883, and the second seed is 0.013 lower, so it is not met. The arm was later found to carry the class-asymmetric form shortcut described below: it calls 47.8, 82.3 and 88.3% of whole-crop-smoothed genuine frames forged, and it ranks only 27.3% of the commercial renders above their originals (seed 1), which may come from the same cue: the commercial smoothing is face-local, the form shown on genuine faces in training. Its low accusation is therefore not evidence that smoothing views suffice; the class-symmetric arm below is.

The genuine-only recipe is an evasion route. P1-smooth lowers the accusation to 0.7, 1.0 and 0.9% (three seeds), ranks the render above its original in 2.6, 6.5 and 5.4% of pairs (AUROC 0.095, seed 1), and has the best Celeb-DF-B numbers of the LoRA arms (frame AUC under the presets 0.887, 0.896 and 0.893 against P0 0.860, 0.883 and 0.883). A check written before any smoothed forgery was scored asks what such a model does to a forgery that a forger smooths (Table S23). Its miss rate goes from 43.6, 40.0 and 37.1% on untreated forgeries to 96.4–100% under the bilateral forms, 75.1–83.6% under the median filter and 50.6–54.4% under the Gaussian blend (miss rises of $+56$ to $+63$, $+38$ to $+40$ and $+11$ to $+14$ points; lower interval bounds of $+49.9$ or more on the bilateral forms), with the forged score falling in 100% of frames and false alarms on smoothed genuine frames at 0.0%. P0 of the third seed rises by $+24.0$, $+14.3$, $+4.3$, $+12.6$ and $+5.6$. The arm with all beautified views on genuine faces only behaves the same ($+46$ to $+51$ on the bilateral forms). The shortcut "smoothed means genuine" removed the accusation and carries over to smoothing forms the views did not contain. The check fails on all three seeds.

The both-class arms carry a shortcut of the opposite sign. A whole-crop bilateral filter on a *genuine* frame is called forged by P2-smooth in 47.8, 82.3 and 88.3% of frames at the three strengths, by OURS, full model in 34.9, 79.8 and 90.1%, by P2 in 3.0, 31.8 and 49.1%, against P0 at 0.8, 16.8 and 39.6% and OURS, forgery head only at 9.9, 35.5 and 45.6% (second seed 2.2, 9.3 and 17.2%). In the training code genuine items receive the face-local scripted smoothing and pre-rendered smoothing frames, while forged items without landmarks receive only the whole-crop bilateral filter, so the form of the smoothing predicts the class. This class-asymmetric augmentation affects every both-class arm and both full-data models.

A class-symmetric view (P2-sym). A smoothed view is drawn the same way wherever landmarks exist: with probability 0.5 the face-local scripted smoothing, otherwise the whole-crop bilateral filter at one of three strengths; forged frames without landmarks receive the whole-crop form; no pre-rendered smoothing frames. Both forms occur in both classes, in proportions that are not identical (genuine 0.5 / 0.5; forged with landmarks 0.33 / 0.67; forged without landmarks 0 / 1). Bars: (a) Smoothing-90 at most 15% at the 5%-FPR rule and the render ranked above its original in 25–75% of pairs; (b) false alarms on smoothed genuine Celeb-DF-v2 frames at most 15% for every form; (c) a miss rise on smoothed forgeries at most 5 points above P0's for every form; two seeds. Bars (a) and (b) are met on both seeds: Smoothing-90 5.9 and 6.1%, the render above its original in 53.7 and 51.8% of pairs, false alarms on smoothed genuine frames 2.4, 5.0, 6.1, 4.5 and 3.6% (seed 1) and 2.2, 4.2, 5.7, 4.5 and 3.0% (seed 2) across the five forms. Bar (c) is not met: the miss rises are $+14.8$, $+14.5$, $+14.6$, $+7.6$ and $+9.1$ (seed 1) and $+19.0$, $+20.6$, $+17.5$, $+9.0$ and $+12.7$ (seed 2), against P0's $+27.0$, $+5.1$, -16.0, $+19.2$ and $+3.0$ and $+26.7$, $+6.1$, -11.8, $+18.0$ and $+8.1$; the arm exceeds the reference by more than 5 points at the mid and strong bilateral strengths on both seeds and under the Gaussian blend on the first (by 4.6 on the second). Part of the reason is the reference: P0's miss rate falls under strong smoothing because P0 accuses everything smoothed (false alarms of 39.6 and 42.9% on smoothed genuine frames at that strength). The arm's own misses still rise by 8–21 points. Read without a threshold, with both classes smoothed, its frame AUC is 0.835, 0.796, 0.767, 0.829 and 0.850 (seed 1) and 0.849, 0.809, 0.796, 0.853 and 0.865 (seed 2) against P0's 0.848, 0.776, 0.747, 0.852 and 0.868 and 0.852, 0.781, 0.736, 0.855 and 0.871: above P0 at the two heavier bilateral strengths, lower by at most 0.023 elsewhere. On untreated data its Celeb-DF-v2 video AUC is 0.919 and 0.934 against P0's 0.938 and 0.938 and its DFD video AUC 0.972 and 0.968 against 0.981 and 0.979. On Celeb-DF-B its frame AUC under the presets is 0.866 and 0.888 against 0.860 and 0.883, its beautification-specific miss rise $+14.2$ and $+10.4$ against $+17.0$ and $+14.2$. On DF40 its video AUC is 0.968 and 0.950 on face swap and 0.871 and 0.796 on reenactment in the FF++ domain, 0.913 and 0.935 and 0.670 and 0.646 in the Celeb-DF domain.

Under Effort's tuning scheme. Declared: the same view, FF++ only, in Effort's scheme from CLIP and as a two-epoch fine-tune of the released Effort checkpoint (learning rate 2×10^{-5}), one seed, bars (a)–(c) against each scheme's

own reference; both pass: scheme-independent; one passes: reported per scheme; neither: the LoRA result stands alone. Effort scheme from CLIP: (b) false alarms on smoothed genuine frames 2.0, 4.2, 6.1, 7.3 and 4.3% (reference 3.6, 35.4, 75.1, 4.8 and 7.6%), met. (a) Smoothing-90 14.5% (reference 56.9%), but the render above its own original in 88.4% of pairs, not met. (c) miss rise $+22.4$, $+25.0$, $+24.9$, $+5.8$ and $+7.2$ against $+19.0$, -11.3, -32.2, $+11.2$ and -2.1, not met. Released Effort fine-tuned for two epochs: Smoothing-90 39.9%, above own original 96.6% (reference 55.3% and 98.5%), (a) not met. Neither passes; the LoRA result stands alone, and the view removes the accusation of the commercial service only under LoRA.

The full-data model with the symmetric view. Declared: the recipe of OURS, full model with the symmetric view (the scripted smoothing of a genuine view is, with probability 0.5, the whole-crop bilateral filter; pre-rendered smoothing frames dropped; everything else unchanged), seed 20260929, checkpoint by OURS, full model's rule. It replaces OURS, full model if (i) whole-crop-smoothed genuine Celeb-DF-v2 false alarms are at most 15% at every strength, (ii) Smoothing-90 is at most 15% with the render above its original in 25–75% of pairs, (iii) untreated Celeb-DF-v2 and DFD video AUC are not lower by more than 0.010, (iv) the explanation flip rate on the three part-forgery editors is not lower by more than 1.0 point, (v) Celeb-DF-B frame AUC under the presets is not lower by more than 0.010. Result: (i) 1.0, 1.6 and 1.7% (was 34.9, 79.8 and 90.1%), met. (ii) 8.1% and 30.0% (was 8.6% and 32.5%), met. (iii) Celeb-DF-v2 0.954 (0.947) and DFD 0.983 (0.978), met. (iv) flip rate on donor transplant, SD-1.5 and SDXL inpainting 92.6, 92.2 and 91.9% (94.0, 93.6 and 93.5%), lower by 1.4, 1.4 and 1.6 points: not met. The ceiling fell by the same amount (92.6, 92.1 and 92.0%; was 94.1, 93.7 and 93.6%), so flip minus ceiling is 0.0, $+0.1$ and -0.1 points against -0.1, -0.1 and -0.1, and IoU_7 is unchanged (0.836, 0.843 and 0.833). The bar was absolute; a bar relative to the ceiling, which we did not declare, would have been met. (v) 0.868 (0.866), with a beautification-specific miss rise of $+10.0$ ($+6.7$) and false alarms on beautified genuine frames of 9.2% (9.5%). Also: validation selection score 0.871 (0.862); untouched originals of the service sent to the beautified class 13.9% (20.8%); frame AUC with both classes smoothed 0.878, 0.831, 0.796, 0.880 and 0.878 (0.803, 0.740, 0.688, 0.819 and 0.855); miss rise on smoothed forgeries $+28.1$, $+33.8$, $+34.1$, $+12.8$ and $+3.8$ (miss rate 43.5% untreated, 77.6% under the strongest bilateral form), where OURS, full model's -25 to -44 came from accusing everything smoothed. By the rule, (iv) fails, so OURS, full model stays and the retrained model is reported next to it as the symmetric-view variant. On DF40 the variant reaches 0.952 and 0.909 on face swap and 0.738 and 0.539 on reenactment (OURS, full model 0.952, 0.905, 0.718 and 0.502).

S18 Smoothing-invariant consistency: the rule, its bars and what it recalibrates

The family of arms of Finding 6 (moved from the main text, Sec. III-F, for space; unchanged):

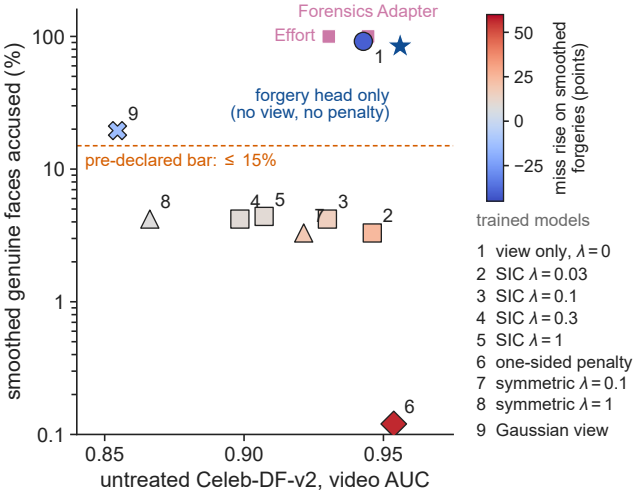


Figure S3: The trade-off traced by the SIC family. Each point is one trained model: untreated Celeb-DF-v2 video AUC against the share of skin-smoothed genuine faces accused (face-local 0.12, log scale); colour gives the rise of the miss rate on the same smoothing applied to forgeries. The star is the model without the rule; squares are the published CLIP detectors. Every penalty weight removes the accusation; smaller weights keep untreated detection and let more smoothed forgeries through.

The rule tested in Finding 6 of the paper follows from the exposure arms (Sec. S17): for every training item x, genuine, beautified or forged, a smoothed copy x_s is drawn by one operator with one distribution of strengths, so that no class can learn the form of the smoothing as a label, and the forged logit of the copy is asked not to move:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}}\big(z_f(x), y\big) + \mathcal{L}_{\text{map}} + \lambda \left(\text{sg}[z_f(x)] - z_f(x_s)\right)^2, \tag{1}$$

where sg stops the gradient: the clean logit is the target, only the smoothed view is pulled, and the clean path is trained by the classification loss alone. The map target of x_s is the footprint of x. With the gradient through both sides (the "without stop-gradient" rows of Table S26) the clean forgery score of a forgery whose cue is texture is pulled down toward its smoothed copy and untreated detection falls further. The operator is face-local bilateral smoothing inside the landmark hull at a random strength, or whole-crop bilateral smoothing where no landmarks are carried; the median filter, the Gaussian blend, the commercial renders and the AR renders are never used in training. The one-sided variant penalises only a rise of the logit; the Gaussian ablation replaces the operator by a Gaussian blur of the same support.

Bars. Written before training (READOUT46): (1) false alarms at most 15% on the face-local smoothed genuine Celeb-DF-v2 frames; (2) at most 15% on the whole-crop bilateral forms and on the commercial Smoothing-90 renders; (3) a miss rise on smoothed forgeries at most 5 points above the arm without views, as written, and (3′) the relaxed form, at most $\max(\text{P0 rise}, 0) + 5$; (4) untreated Celeb-DF-v2 and DFD video AUC not lower than the model without the rule by more than 0.010; (5) Celeb-DF-B AUC under the presets not below the main model; (6) DF40 reenactment video AUC not lower by more than 0.030; a $\lambda = 0$ control; two seeds. Point estimates decide. Bar (3) as written is met by no member; the "passed" count of Table V counts bars 1, 2, 3′, 4, 5 and 6.

Table S24: Bars met by every trained member of the family (READOUT46). Bar numbers as in the text; "passed" counts bars 1, 2, 3′, 4, 5, 6.

Arm	seed	1 face-local FA	2 whole-crop and S-90	3 as written	3′ relaxed	4 untreated video	5 Celeb-DF-B AUC	6 DF40 reenactment	passed
SIC factorised (primary)	1	✓	✓	–	–	–	–	–	2/6
SIC binary (secondary)	1	✓	✓	–	–	–	–	–	2/6
SIC binary (secondary)	2	✓	✓	–	–	–	–	–	2/6
SIC binary, lambda 0 control	1	–	✓	–	–	–	✓	✓	3/6
SIC binary, lambda 0.3	1	✓	✓	–	–	–	–	–	2/6
SIC binary, lambda 0.1	1	✓	✓	–	–	–	✓	–	3/6
SIC binary, stop-gradient	1	✓	✓	–	–	–	–	–	2/6
SIC binary, stop-gradient	2	✓	✓	–	–	–	–	–	2/6
FF++-only SIC binary, stop-gradient	1	✓	✓	–	–	–	–	–	2/6
SIC-sg binary, lambda 0.3	1	✓	✓	–	–	–	–	–	2/6
SIC-sg binary, lambda 0.1	1	✓	✓	–	–	–	✓	–	3/6
SIC-sg binary, lambda 0.1	2	✓	✓	–	–	–	–	–	2/6
SIC-sg binary, Gaussian operator ablation	1	–	✓	–	–	–	–	–	1/6
SIC-sg binary, Gaussian operator ablation	2	✓	✓	–	–	–	–	–	2/6
SIC-h binary, one-sided consistency	1	✓	✓	–	–	✓	✓	–	4/6
SIC-h binary, one-sided consistency	2	✓	✓	–	–	–	–	–	2/6
SIC-sg binary, lambda 0.03	1	✓	✓	–	–	✓	✓	–	4/6
SIC-sg binary, lambda 0.03	2	✓	✓	–	–	–	–	–	2/6

Table S25: The full family, including the variants without the stop-gradient (symmetric penalty) and the FF++-only member. Columns as in Table V.

Model	seed	smoothed genuine accused (%) fl 0.06	fl 0.12	bil 15	S-90 acc. %	Δmiss fl 0.12	untreated video AUC CDFv2	DFD	Celeb-DF-B AUC	DF40 FR ff	bars passed
forgery head only, no rule	1	51.6	84.5	35.5	9.1	−34.0	0.956	0.987	0.866	0.842	–
symmetric view only ($\lambda = 0$)	1	53.9	91.8	1.6	11.9	−40.1	0.943	0.975	0.891	0.835	3/6
SIC, $\lambda = 0.03$	1	5.0	3.3	9.6	11.2	24.4	0.946	0.979	0.886	0.750	4/6
SIC, $\lambda = 0.03$	2	3.8	3.5	8.8	15.0	20.4	0.933	0.975	0.876	0.777	2/6
SIC, $\lambda = 0.1$	1	6.5	4.2	9.9	10.5	15.8	0.930	0.980	0.871	0.774	3/6
SIC, $\lambda = 0.1$	2	4.6	4.6	8.5	8.1	17.6	0.929	0.974	0.868	0.771	2/6
SIC, $\lambda = 0.3$	1	6.0	4.2	8.1	12.8	9.5	0.898	0.971	0.840	0.692	2/6
SIC, $\lambda = 1$	1	5.8	4.4	5.8	2.1	9.7	0.907	0.986	0.835	0.768	2/6
SIC, $\lambda = 1$	2	6.2	5.5	6.3	12.3	6.1	0.844	0.968	0.804	0.746	2/6
one-sided penalty	1	0.1	0.1	0.0	9.2	56.5	0.954	0.980	0.880	0.794	4/6
one-sided penalty	2	0.1	0.0	0.0	10.4	47.3	0.929	0.975	0.860	0.775	2/6
Gaussian view (ablation)	1	5.8	19.5	1.1	3.5	−13.5	0.855	0.971	0.819	0.782	1/6
Gaussian view (ablation)	2	5.9	14.5	1.3	8.7	−9.3	0.848	0.960	0.821	0.784	2/6
SIC, $\lambda = 1$, FF++ only	1	7.4	6.8	5.3	7.0	4.8	0.856	0.979	0.829	0.674	2/6
symmetric penalty, $\lambda = 0.1$	1	4.3	3.3	7.1	9.5	18.6	0.921	0.974	0.882	0.760	3/6
symmetric penalty, $\lambda = 0.3$	1	4.7	4.6	8.1	9.9	12.4	0.900	0.966	0.860	0.755	2/6
symmetric penalty, $\lambda = 1$	1	4.9	4.2	7.0	7.1	7.8	0.866	0.956	0.837	0.702	2/6
symmetric penalty, $\lambda = 1$	2	4.7	5.3	6.9	7.6	6.3	0.858	0.955	0.829	0.712	2/6
symmetric penalty, full model, $\lambda = 1$	1	4.5	4.3	7.2	7.6	10.2	0.876	0.968	0.845	0.663	2/6
smoothing expert (trained on smoothed copies only; routed model B)	1	10.8	16.4	93.5	12.5	−11.2	0.904	0.978	0.836	0.529	0/6
smoothing expert (trained on smoothed copies only; routed model B)	2	11.7	13.6	76.6	12.8	−6.3	0.880	–	–	0.468	2/4
FF++-only P0 without Deepfakes and NeuralTextures	1	95.5	100.0	17.1	33.5	−45.0	0.918	0.984	0.868	0.631	2/6

Recalibration control. Table S26 reads every model with both classes smoothed by the same form and the threshold re-set at 5% FPR on the smoothed genuine frames of that form (an oracle): the cell is the AUC between smoothed genuine and smoothed forged frames and the miss rate at that threshold. The $\lambda = 0$ model (symmetric view, no penalty) read this way has a both-classes-smoothed AUC of 0.832 and 0.764 (bilateral 15, face-local 0.12) against 0.790 and 0.778 for SIC at $\lambda = 1$ and 0.802 and 0.778 at $\lambda = 0.1$, and a recalibrated miss of 57.4 and 76.7% against 64–73% for SIC on two seeds; this is the basis of the paper's reading of SIC as a smoothing-conditional recalibration. Every published detector loses AUC with both classes smoothed (0.06–0.28 at face-local 0.12 for the sixteen above chance), and the $\lambda = 0$ model loses 0.061 and 0.129 from its untreated 0.893.

Table S26: Both classes smoothed, threshold re-set at 5% FPR on the same condition's genuine frames (oracle recalibration). Cell: AUC / miss rate (%). Celeb-DF-v2 test, frames with landmarks (RECAL_MISS).

Model	untreated	bilateral 15	bilateral 21	face-local 0.06	face-local 0.12
OURS, forgery head only, seed 1	0.909 / 36.5	0.805 / 63.1	0.754 / 67.8	0.886 / 47.0	0.755 / 73.7
OURS, forgery head only, seed 2	0.917 / 35.4	0.843 / 52.0	0.810 / 59.0	0.879 / 48.8	0.748 / 75.4
symmetric view only ($\lambda = 0$)	0.893 / 40.7	0.832 / 57.4	0.798 / 64.5	0.870 / 53.2	0.764 / 76.7
SIC without stop-gradient, $\lambda = 1$, seed 1	0.813 / 62.9	0.756 / 69.1	0.752 / 70.7	0.790 / 64.9	0.749 / 68.9
SIC without stop-gradient, $\lambda = 1$, seed 2	0.805 / 65.2	0.751 / 71.6	0.742 / 71.8	0.783 / 66.0	0.733 / 72.6
SIC, $\lambda = 1$, seed 1	0.841 / 55.5	0.790 / 64.5	0.791 / 63.6	0.825 / 58.0	0.778 / 64.0
SIC, $\lambda = 1$, seed 2	0.782 / 66.0	0.733 / 73.0	0.731 / 73.5	0.769 / 69.1	0.720 / 72.8
SIC, $\lambda = 0.3$	0.829 / 59.1	0.765 / 68.5	0.756 / 71.7	0.819 / 59.3	0.759 / 66.4
SIC, $\lambda = 0.1$	0.864 / 47.6	0.802 / 62.8	0.795 / 61.3	0.841 / 53.5	0.778 / 61.0
one-sided penalty, seed 1	0.892 / 42.9	0.727 / 71.9	0.691 / 76.4	0.667 / 71.9	0.603 / 77.4
one-sided penalty, seed 2	0.862 / 52.3	0.738 / 72.6	0.714 / 76.0	0.654 / 77.7	0.606 / 81.5
Gaussian view (ablation)	0.802 / 65.4	0.703 / 75.6	0.694 / 77.3	0.760 / 69.1	0.705 / 72.6
Xception	0.717 / 77.7	0.544 / 93.3	0.487 / 96.1	0.669 / 83.8	0.580 / 90.7
EfficientNet-B4	0.746 / 76.7	0.590 / 95.2	0.553 / 94.9	0.679 / 90.7	0.531 / 97.0
F3Net	0.712 / 82.0	0.611 / 92.4	0.577 / 94.2	0.697 / 85.6	0.652 / 95.5
SPSL	0.703 / 79.3	0.561 / 92.2	0.533 / 92.1	0.652 / 83.2	0.631 / 84.2
UCF	0.742 / 74.2	0.490 / 95.4	0.423 / 96.1	0.673 / 89.6	0.535 / 94.8
RECCE	0.689 / 84.6	0.552 / 95.2	0.545 / 95.4	0.688 / 84.7	0.607 / 92.0
CORE	0.720 / 82.3	0.576 / 91.3	0.596 / 93.0	0.678 / 82.9	0.646 / 100.0
SRM	0.719 / 83.7	0.486 / 97.4	0.504 / 96.4	0.590 / 92.9	0.513 / 94.7
SBI	0.815 / 62.3	0.644 / 88.5	0.531 / 94.6	0.660 / 89.9	0.577 / 93.4
LAA-Net	0.847 / 56.1	0.651 / 84.9	0.582 / 90.3	0.603 / 89.5	0.566 / 92.3
CADDM	0.832 / 57.3	0.610 / 90.9	0.590 / 93.1	0.733 / 74.9	0.572 / 91.7
Effort	0.876 / 41.3	0.785 / 67.7	0.748 / 76.6	0.749 / 81.1	0.658 / 84.0
Forensics Adapter	0.883 / 42.9	0.705 / 82.5	0.668 / 87.6	0.705 / 82.3	0.649 / 87.6
GenD	0.887 / 39.9	0.771 / 80.1	0.712 / 83.5	0.677 / 90.9	0.632 / 90.8
DFD-HR	0.869 / 45.7	0.751 / 72.0	0.713 / 75.5	0.701 / 86.5	0.601 / 93.5
UnivFD	0.663 / 83.4	0.524 / 94.9	0.511 / 96.0	0.689 / 83.5	0.579 / 97.6
NPR	0.527 / 93.6	0.500 / 100.0	0.500 / 100.0	0.538 / 100.0	0.521 / 100.0

Routed expert. A second model B trained only on smoothed copies (x_s of every class) is selected by the smoothing score p_s of the three-way model (route to B when p_s exceeds the FF++-validation threshold); A is the forgery-head-only model of the same seed. Table S27 gives seed 20260930. Routing clears the accusation on the face-local forms (51.2 and 79.9% to 2.1 and 1.3%), raises the miss on the same smoothed forgeries by 40.6 and 46.0 points, and B's both-classes-smoothed AUC is below A's on seven of the eight forms. On the commercial renders the routed decision accuses 0.2% of Smoothing-90 renders (A: 6.5%) and 4.8% of the originals; 99.4% of the renders and 0.9% of the originals are routed. Seed 20260929 (SCE_S1) repeats the pattern: face-local 51.6 and 84.5% to 0.9 and 1.3%, miss rise 41.7 and 44.4 points, B below A on seven of the eight forms (face-local 0.12 the exception on both seeds), Smoothing-90 renders accused 0.1% (A: 9.1%).

Table S27: Routed expert (SCE), seed 20260930. Share of frames routed to the smoothing expert B, false alarms and misses of A alone and of the routed decision at the 5%-FPR rule, and the AUC with both classes smoothed of A and of B (SCE_S2).

Form	routed to B (%)	FA, A → routed [CI]	MISS, A → routed	miss rise, A → routed	AUC both smoothed, A / B
untreated	0.0	5.0 → 5.0 [3.7, 6.2]	35.4 → 35.4	0.0 → 0.0	0.917 / 0.8129
bil9	99.9	2.2 → 6.8 [4.6, 8.3]	57.8 → 56.7	22.4 → 21.3	0.8785 / 0.8014
bil15	100.0	9.3 → 7.8 [5.8, 9.6]	41.7 → 65.3	6.3 → 29.8	0.843 / 0.7518
bil21	100.0	17.2 → 6.2 [4.5, 7.7]	37.6 → 69.2	2.2 → 33.7	0.81 / 0.7443
median5	74.4	3.4 → 5.5 [3.9, 7.0]	49.0 → 56.8	13.6 → 21.4	0.8825 / 0.7511
gblend	3.6	3.6 → 3.4 [2.3, 4.5]	43.7 → 46.0	8.4 → 10.6	0.8996 / 0.8009
fl06	99.8	51.2 → 2.1 [0.8, 3.0]	4.6 → 76.0	−30.8 → 40.6	0.879 / 0.816
fl12	100.0	79.9 → 1.3 [0.5, 2.0]	3.2 → 81.4	−32.2 → 46.0	0.7484 / 0.7697

S19 A smoothing-conditional threshold, every detector

Table S28 gives the rule of Finding 5 (Table IV) for every detector: fake iff $s > t_0 + \Delta(b)$, with t_0 the 95th percentile of untreated genuine frames and Δ learnt on FF++ validation genuine renders only, binned by the skin-smoothing presence probability b of the three-way model (bins at 0.1, 0.5, 0.9). Cells read fixed → conditional. The last column is the shift learnt for the top bin. The rule lowers the accusation of smoothed genuine faces for every detector that is sensitive to b and raises the miss on smoothed forgeries by the same mechanism; NPR, whose scores saturate, is unchanged.

Table S28: Beautification-conditional threshold (BCT) for every detector. FA on untreated, face-local 0.06 / 0.12 and bilateral 15 smoothed genuine Celeb-DF-v2 frames and on the commercial Smoothing-90 renders; MISS on face-local 0.12 and bilateral 15 smoothed forgeries and on untreated forgeries (%), fixed → conditional (BCT).

Detector	untreated FA	fl 0.06 FA	fl 0.12 FA	bil 15 FA	S-90 accused	fl 0.12 MISS	bil 15 MISS	untreated MISS	Δ (top bin)
Xception	5.0 → 5.0	89.2 → 64.4	99.3 → 92.4	76.7 → 22.4	51.6 → 0.0	0.2 → 4.2	18.1 → 72.7	77.7 → 79.1	0.1308
EfficientNet-B4	5.0 → 5.0	64.3 → 51.5	95.0 → 90.4	79.7 → 67.8	36.3 → 0.0	1.1 → 2.9	9.1 → 20.4	76.7 → 77.3	0.077
F3Net	5.0 → 5.0	86.3 → 81.0	99.6 → 99.3	98.5 → 94.6	50.6 → 0.0	0.0 → 0.0	0.1 → 0.7	82.0 → 82.9	0.0753
SPSL	5.0 → 5.0	70.3 → 28.4	92.1 → 65.5	63.1 → 8.4	51.2 → 0.9	3.6 → 19.6	29.9 → 86.8	79.3 → 80.4	0.0551
UCF	5.0 → 5.0	89.3 → 88.8	98.4 → 98.0	24.3 → 22.6	41.4 → 0.0	0.5 → 0.8	77.5 → 79.2	74.2 → 74.3	0.0066
RECCE	5.0 → 5.0	80.7 → 22.7	97.8 → 45.3	91.8 → 2.0	36.8 → 0.0	0.5 → 39.7	3.4 → 97.5	84.6 → 86.1	0.0582
CORE	5.0 → 5.0	85.0 → 53.2	99.7 → 97.6	99.3 → 88.4	54.8 → 0.0	0.1 → 0.8	0.1 → 3.8	82.3 → 84.1	0.1464
SRM	5.0 → 5.0	81.0 → 12.8	91.8 → 9.4	74.0 → 5.4	62.8 → 17.3	5.7 → 90.9	25.8 → 97.1	83.7 → 84.8	0.0808
SBI	5.0 → 5.0	96.7 → 0.0	100.0 → 0.0	72.3 → 0.0	37.3 → 0.0	0.0 → 100.0	10.7 → 100.0	62.3 → 66.4	0.2027
UnivFD	5.0 → 5.0	13.9 → 0.9	55.2 → 11.1	40.8 → 5.8	16.6 → 1.0	28.8 → 90.8	56.1 → 93.8	83.4 → 83.5	0.7776
NPR	5.0 → 5.0	68.0 → 68.0	77.7 → 77.7	100.0 → 100.0	0.0 → 0.0	18.7 → 18.7	0.0 → 0.0	93.6 → 93.6	0.0
Effort	5.0 → 4.9	87.8 → 0.0	99.9 → 0.0	27.1 → 0.0	43.2 → 20.5	0.0 → 100.0	32.2 → 100.0	41.3 → 47.0	0.2118
Forensics Adapter	5.0 → 4.9	95.8 → 0.1	100.0 → 0.0	59.9 → 0.0	39.7 → 31.4	0.0 → 100.0	13.3 → 100.0	42.9 → 48.7	0.1975
GenD	5.0 → 5.0	96.1 → 3.9	100.0 → 11.7	81.7 → 0.6	68.7 → 32.4	0.0 → 78.3	1.5 → 93.7	39.9 → 42.9	0.1187
DFD-HR	5.0 → 5.0	99.3 → 0.4	100.0 → 0.6	79.8 → 0.0	64.9 → 36.8	0.0 → 99.3	4.4 → 100.0	45.7 → 47.6	0.3084
CADDM	5.0 → 5.0	88.5 → 0.0	99.9 → 0.0	61.4 → 0.0	6.3 → 0.0	0.0 → 100.0	24.3 → 100.0	57.3 → 60.6	0.1479
LAA-Net	5.0 → 4.8	100.0 → 98.8	100.0 → 100.0	92.1 → 61.1	17.2 → 0.0	0.0 → 0.0	1.6 → 20.9	56.1 → 57.8	0.1241
OURS, forgery head only	5.0 → 5.0	51.6 → 0.0	84.5 → 0.1	35.5 → 0.0	9.1 → 4.9	2.5 → 100.0	21.0 → 100.0	36.5 → 36.5	0.0125
OURS (three-way)	5.0 → 4.8	5.8 → 0.0	44.5 → 0.0	79.8 → 0.0	8.6 → 0.9	32.9 → 100.0	4.7 → 100.0	47.0 → 53.2	0.3299
SIC, $\lambda = 1$	5.0 → 5.0	5.8 → 4.2	4.4 → 3.0	5.8 → 3.9	2.1 → 0.7	65.3 → 69.6	61.2 → 68.1	55.5 → 55.6	0.0849
SIC, $\lambda = 0.1$	5.0 → 5.0	6.5 → 0.0	4.2 → 0.0	9.9 → 0.0	10.5 → 3.3	63.4 → 99.9	51.4 → 100.0	47.6 → 47.9	0.0572

S20 The face against its context, and a test-time rule that does not work

The band-energy analysis of the main text says that the cue is face-local and edge-preserving. A second probe asks whether the cue is the line where the smoothed face meets the untouched background. Pre-registered on 2026-10-23, the same bilateral strength as the face-local 0.12 form is applied to the Celeb-DF-v2 test frames of Sec. S9 in four face-local forms whose only difference is the edge treatment (hard edge, then feathered over 1, 5 and 15% of the crop width), to the whole crop, and to a band at the face edge alone, one tenth of the jaw width wide, with the face interior untouched. Every detector keeps its own threshold, set at 5% false alarms on the untreated frames. Fig. S4 gives the three forms that decide the question. The ring alone accuses a median of 12% of the frames (7–51%) for the thirteen face detectors other than CADDM and LAA-Net, against 92–100% for the face-local form (prediction "ring reaches 70% of face-local" holds for 1 of 15 face detectors, bar 10 of 15); the four CLIP detectors and SBI are the upper half of that range (22–51%). Feathering the edge lowers the accusation monotonically for 14 of 15, so the edge contributes, but it is not the cue. Smoothing the whole crop accuses less than smoothing the face alone for 11 of 15 (by 10 points or more), so the detectors read a face that is smoother than its surroundings. CADDM (ring 94%) and LAA-Net (ring 65%) are the two detectors that read the edge; both are trained on blending boundaries. Our three-way model runs the other way (face-local 45%, whole crop 81%): it has seen whole-crop smoothing of genuine frames in training as a beautified view and reads it as a class cue, the shortcut described in Sec. S17.

This mechanism suggests a zero-training rule: score the image and a whole-crop-smoothed copy of it, and call the image forged only if both exceed their thresholds, since the copy removes the face-context contrast of a beautified genuine face while a forgery should keep evidence that does not depend on it. Pre-registered on 2026-10-27 with two predictions. The first holds at its bar: the joint rule accuses fewer than 30% of face-locally smoothed genuine frames for 10 of 15 face detectors (single score: 92–100%). The second, which decides whether the rule is more than a threshold, fails for every detector: at a matched 5% false-alarm rate on the same smoothed genuine frames, the joint rule misses smoothed forgeries within a few points of the recalibrated single score (Effort 86.5 vs 84.0%, Forensics Adapter 89.2 vs 87.6, GenD 90.2 vs 90.8, DFD-HR 90.7 vs 93.5, Xception 93.8 vs 90.7; 0 of 15 better by 10 points, kill bar 5 of 15), and the second gate raises the miss rate on untreated forgeries by 20 to 35 points (Effort 41 to 77%, SBI 62 to 96%). Once a smoothed genuine face and a smoothed forgery both pass the first threshold, the score on the context-matched copy carries no further separation between them. This is the fourth independent closure of the same bound (threshold, routing, consistency training, context matching).

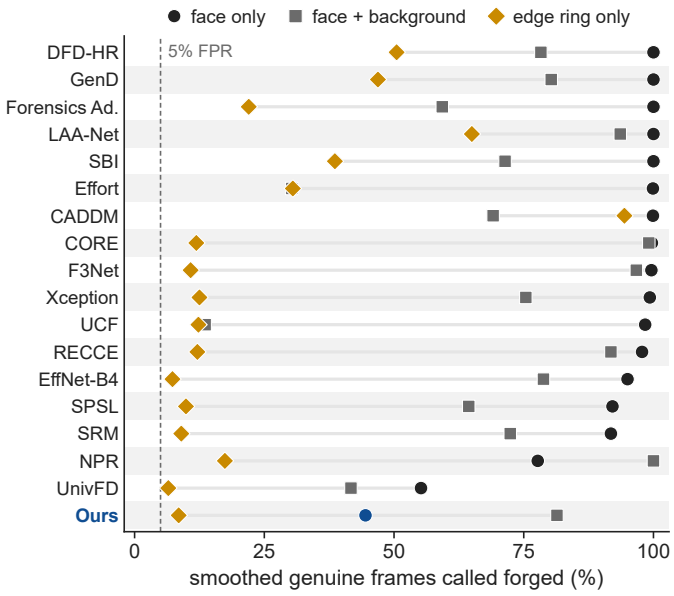


Figure S4: Smoothed genuine Celeb-DF-v2 test frames called forged at each detector's own 5% FPR threshold, with the same bilateral strength applied to the face only, to the whole crop, and to a band at the face edge. Rows sorted by the face-only share. The band alone accuses far less than the face for every detector except CADDM and LAA-Net; the whole crop accuses less than the face alone for most detectors and more for ours (BDRY38).

S21 A second beauty engine: GPUPixel

Pre-registered on 2026-10-21, the skin smoothing of the open-source GPUPixel beauty engine (BeautyFaceFilter, commit ef552bf8), ported to numpy from its shaders with 8-bit intermediate buffers, is applied to the whole crop of both classes of the Celeb-DF-v2 test frames of Sec. S9 at blur strengths 0.3, 0.6 and 0.9 (levels 3, 6 and 9 of the demo slider); level 0 passes the frame through the same pipeline without smoothing and is the null form. No design decision of this study touched this source. Three predictions were written: (1) at level 9 more than 15% of genuine frames are accused for at least 10 of 15 face detectors, which holds for 14 (SRM 12.5% is the exception; level 0 gives 2.4–4.9%); (2) the AUC with both classes smoothed falls monotonically with the level for at least 12 of 15, which holds for 11 and fails; (3) the ranking of the detectors by accusation agrees with the scripted bilateral form (Spearman at least 0.5), which gives 0.243 and fails. The accusation replicates on an untouched engine; the ordering of the detectors does not. Our forgery-head arms accuse 17.1–24.6% at level 9. The port is not the compiled library, so we do not claim byte equivalence (GPX38).

S22 Training without the texture-poor FF++ families

Deepfakes and NeuralTextures forgeries have less high-frequency energy inside the face than their source frames (ratio -1.09 and -0.65; FFPP_BAND). Pre-registered on 2026-10-21, an FF++-only arm with the P0 recipe was trained on Face2Face and FaceSwap forgeries alone. It accuses face-local smoothing at 95.5% (strength 0.06) and 100% (0.12), against 95.8 and 100% for P0 on both seeds, with an untreated Celeb-DF-v2 AUC of 0.857 (P0: 0.883 and 0.892). The predicted fall of at least 15 points did not occur, so the texture-poor families do not teach the cue and the paper attributes no origin to them.

S23 Identity as a reference channel

The probes are the Celeb-DF-v2 test frames of Sec. S9 restricted to Celeb-real and Celeb-synthesis (3,504 frames: 836 genuine, 2,668 forged), untreated and under four smoothing forms; face detection fails on none. The identity score is the ArcFace distance between the probe and a reference set of the claimed person's other videos (54 identities, 1,080 frames, none from the test videos). Intervals are bootstraps over claimed identities of the AUC change against untreated.

Table S29 gives the AUC and the error rates at the untreated 5%-FPR threshold held fixed; the fusion row (mean of the two quantile scores) is dominated by the identity distance alone at every condition and is not reported as a method in the paper. Table S30 gives ICT and ICT-Ref [25] on the same probes with the released weights: the main ICT-Ref row uses our reference set through ICT's own alignment and $k = 1$ nearest neighbour over the pool, as in the released evaluation; the released reference file (2,420 entries of unknown provenance) and a claimed-identity pool are listed as secondary readings. Their AUC falls by 0.08–0.12 under the three stronger forms with intervals wholly below -0.05, and their failure direction is the opposite of the texture detectors': the forged median score falls, misses rise and the false alarm does not. ID-Reveal was not run, because its temporal network reads 3DMM coefficients of consecutive frames that our per-frame probes do not provide.

Table S29: Identity distance against texture detectors on the same probes (ID_REF). Top: AUC, with the 95% identity-bootstrap interval of the change against untreated. Bottom: false alarm / miss (%) at the untreated 5%-FPR threshold.

Score	untreated	bilateral 15	bilateral 21	face-local 0.06	face-local 0.12
identity (ArcFace distance)	0.9962	0.9839 [−0.0219, −0.0053]	0.9629 [−0.0545, −0.0166]	0.9946 [−0.0034, −0.0003]	0.9815 [−0.0264, −0.0053]
Ours, forgery head only	0.9244	0.8229 [−0.1238, −0.0812]	0.7776 [−0.1755, −0.1199]	0.8869 [−0.0686, −0.0105]	0.7507 [−0.2135, −0.135]
Effort	0.8966	0.8124 [−0.1085, −0.0588]	0.7795 [−0.1476, −0.0862]	0.7626 [−0.1824, −0.0968]	0.663 [−0.2843, −0.1912]
GenD	0.904	0.8042 [−0.1348, −0.0646]	0.7343 [−0.2101, −0.1273]	0.6645 [−0.2918, −0.1872]	0.6055 [−0.3478, −0.2477]
SBI	0.8323	0.6997 [−0.1831, −0.0918]	0.5764 [−0.3161, −0.2004]	0.6376 [−0.2623, −0.1345]	0.5726 [−0.3371, −0.1948]
fusion Ours, forgery head only + identity	0.9895	0.9572 [−0.0454, −0.0206]	0.9343 [−0.0751, −0.0379]	0.9828 [−0.0149, 0.0005]	0.943 [−0.0637, −0.0322]

Score	untreated	bilateral 15	bilateral 21	face-local 0.06	face-local 0.12
identity (ArcFace distance)	5.0 / 0.1	19.4 / 0.1	40.0 / 0.0	6.3 / 0.2	21.3 / 0.3
Ours, forgery head only	5.0 / 33.8	43.2 / 14.4	53.5 / 14.9	56.6 / 3.8	88.0 / 1.8
Effort	5.0 / 33.4	41.5 / 16.8	66.5 / 5.3	94.9 / 0.1	100.0 / 0.0
GenD	5.0 / 30.6	88.0 / 0.6	96.8 / 0.1	98.8 / 0.0	100.0 / 0.0
SBI	5.0 / 55.0	74.2 / 5.7	92.6 / 1.2	98.2 / 0.1	100.0 / 0.0
fusion Ours, forgery head only + identity	5.0 / 5.7	14.8 / 9.4	20.6 / 10.9	7.8 / 6.0	22.6 / 8.7

Table S30: ICT and ICT-Ref with released weights on the same probes (ICT_PROBE). Top: AUC, change against untreated in parentheses, with its 95% identity-bootstrap interval. Bottom: false alarm / miss (%) at the untreated 5%-FPR threshold.

Score	untreated	bilateral 15	bilateral 21	face-local 0.06	face-local 0.12
ICT	0.7898	0.6789 (−0.1109) [−0.1457, −0.0806]	0.6664 (−0.1234) [−0.1695, −0.0831]	0.7518 (−0.0380) [−0.0513, −0.0259]	0.6983 (−0.0915) [−0.126, −0.0604]
ICT-Ref (our reference set)	0.863	0.7811 (−0.0819) [−0.1068, −0.0599]	0.7546 (−0.1084) [−0.1462, −0.0686]	0.8355 (−0.0276) [−0.0364, −0.0197]	0.7828 (−0.0803) [−0.102, −0.0568]
ICT-Ref (released reference)	0.9325	0.8804 (−0.0521) [−0.0684, −0.0362]	0.8402 (−0.0923) [−0.119, −0.0649]	0.9189 (−0.0136) [−0.019, −0.0079]	0.8881 (−0.0444) [−0.0604, −0.0275]
ICT-Ref (claimed-identity reference)	0.793	0.7111 (−0.0819) [−0.1118, −0.0558]	0.7024 (−0.0906) [−0.1318, −0.0501]	0.7617 (−0.0313) [−0.0399, −0.0227]	0.7133 (−0.0797) [−0.1082, −0.0532]
identity (ArcFace distance)	0.9962	0.9839 (−0.0123) [−0.0219, −0.0053]	0.9629 (−0.0333) [−0.0545, −0.0166]	0.9946 (−0.0015) [−0.0034, −0.0003]	0.9815 (−0.0147) [−0.0264, −0.0053]
Ours, forgery head only	0.9244	0.8229 (−0.1016) [−0.1238, −0.0812]	0.7776 (−0.1468) [−0.1755, −0.1199]	0.8869 (−0.0376) [−0.0686, −0.0105]	0.7507 (−0.1737) [−0.2135, −0.135]
Effort	0.8966	0.8124 (−0.0841) [−0.1085, −0.0588]	0.7795 (−0.1170) [−0.1476, −0.0862]	0.7626 (−0.1339) [−0.1824, −0.0968]	0.663 (−0.2335) [−0.2843, −0.1912]
GenD	0.904	0.8042 (−0.0997) [−0.1348, −0.0646]	0.7343 (−0.1697) [−0.2101, −0.1273]	0.6645 (−0.2395) [−0.2918, −0.1872]	0.6055 (−0.2985) [−0.3478, −0.2477]
SBI	0.8323	0.6997 (−0.1326) [−0.1831, −0.0918]	0.5764 (−0.2559) [−0.3161, −0.2004]	0.6376 (−0.1947) [−0.2623, −0.1345]	0.5726 (−0.2597) [−0.3371, −0.1948]

Score	untreated	bilateral 15	bilateral 21	face-local 0.06	face-local 0.12
ICT	5.0 / 65.3	2.3 / 87.8	2.5 / 85.3	3.7 / 73.3	4.7 / 75.7
ICT-Ref (our reference set)	5.0 / 46.6	2.9 / 70.4	7.2 / 64.1	4.1 / 54.6	5.5 / 56.5
ICT-Ref (released reference)	5.0 / 32.3	4.7 / 52.7	12.3 / 39.8	4.9 / 40.0	7.1 / 39.1
ICT-Ref (claimed-identity reference)	5.0 / 62.9	3.1 / 80.8	5.4 / 72.5	4.3 / 69.8	5.4 / 69.5
identity (ArcFace distance)	5.0 / 0.1	19.4 / 0.1	40.0 / 0.0	6.3 / 0.2	21.3 / 0.3
Ours, forgery head only	5.0 / 33.8	43.2 / 14.4	53.5 / 14.9	56.6 / 3.8	88.0 / 1.8
Effort	5.0 / 33.4	41.5 / 16.8	66.5 / 5.3	94.9 / 0.1	100.0 / 0.0
GenD	5.0 / 30.6	88.0 / 0.6	96.8 / 0.1	98.8 / 0.0	100.0 / 0.0
SBI	5.0 / 55.0	74.2 / 5.7	92.6 / 1.2	98.2 / 0.1	100.0 / 0.0

S24 Identity under real app presets (Celeb-DF-B)

Pre-registered on 2026-10-28 before any score was read. The probes are the 32 standardised frames per video of the six Celeb-DF-B folders (43,363 frames; face detection fails on none of the genuine frames). The reference of each claimed person is the normalised mean ArcFace (buffalo_l) embedding of five frames from each of up to four Celeb-DF-v2 Celeb-real videos of that person, excluding the 231 genuine videos of Celeb-DF-B; the identity score is one minus the cosine similarity between the probe and that reference. Texture scores are those of Table II. Fusion is the mean of the two scores after each is mapped to its quantile among the untreated genuine frames; the gate variant replaces the fused score by $\min(\text{fused}, 0.5)$ whenever identity alone calls the frame genuine. All thresholds accuse 5% of the untreated genuine frames unless stated, and beautification-only changes subtract the change under re-encoding, with 95% video-cluster bootstrap intervals. Predictions: (1) the identity video AUC falls by at most 0.03 under the presets (kill above 0.05): holds, no fall;

(2) at a threshold re-set to 5% FA on the beautified genuine frames, identity misses at most 15% and every texture detector at least 40%: holds (0.6 against 46.2–72.2%); (3) both rank-mean fusions keep the beautification-only FA rise within 10 points and the miss rise within 5: fails, Effort plus identity raises the miss by 8.5 [5.4, 11.5]; (4) identity's beautification-only FA rise is at least 10 points: fails narrowly, 8.5 [6.4, 10.6]. The gate variant passes (3) for every pair because its misses equal identity's (CDFB_ID).

Table S31: Identity and texture detectors on Celeb-DF-B. vAUC: video AUC untreated / beautified / re-encoded. ΔFA, ΔMISS: beautification-only change at the untreated 5%-FPR threshold (points, 95% video-bootstrap). Last column: miss on beautified forgeries at the threshold that accuses 5% of the beautified genuine frames.

Score	vAUC untr. / beaut. / re-enc.	ΔFA beauty only	ΔMISS beauty only	MISS at 5% FA on beautified
Identity (ArcFace + reference)	1.000 / 1.000 / 1.000	+8.5 [+6.4, +10.6]	−0.3 [−0.6, −0.1]	0.6
Effort	0.950 / 0.850 / 0.930	+2.1 [+0.7, +3.6]	+11.0 [+7.5, +14.3]	51.9
Forensics Adapter	0.960 / 0.890 / 0.950	+0.1 [−1.9, +2.1]	+15.2 [+11.8, +18.5]	54.4
GenD	0.970 / 0.850 / 0.950	+3.6 [+1.3, +5.9]	+12.0 [+8.3, +15.5]	61.9
DFD-HR	0.980 / 0.890 / 0.960	+9.8 [+7.2, +12.6]	+3.6 [+0.2, +6.8]	53.2
SBI	0.920 / 0.800 / 0.910	−0.6 [−1.7, +0.3]	+21.8 [+17.5, +25.9]	69.6
CADDM	0.960 / 0.830 / 0.930	+9.9 [+8.0, +12.0]	−2.9 [−5.9, +0.1]	72.1
Xception	0.870 / 0.770 / 0.860	−1.7 [−3.2, −0.3]	+10.9 [+7.5, +14.3]	72.2
OURS, seed 1	0.980 / 0.920 / 0.970	+4.9 [+2.8, +7.2]	+6.7 [+3.9, +9.4]	51.1
OURS, seed 2	0.980 / 0.940 / 0.980	+2.7 [+1.5, +4.0]	+8.3 [+5.6, +11.0]	46.2
Fusion, OURS + identity	1.000 / 0.980 / 1.000	+7.7 [+5.9, +9.5]	+2.9 [+1.2, +4.7]	22.7
Fusion, Effort + identity	0.990 / 0.950 / 0.990	+6.2 [+4.2, +8.2]	+8.5 [+5.4, +11.5]	31.5
Fusion, Forensics Adapter + identity	0.990 / 0.960 / 0.990	+5.3 [+3.3, +7.4]	+10.8 [+8.2, +13.4]	28.9
Fusion, GenD + identity	0.990 / 0.950 / 0.990	+6.6 [+4.7, +8.4]	+7.2 [+4.7, +9.6]	29.9
Gate, OURS + identity	1.000 / 0.990 / 1.000	+8.2 [+6.2, +10.3]	−0.3 [−0.6, −0.1]	13.9
Gate, Effort + identity	1.000 / 0.990 / 1.000	+8.0 [+6.1, +10.0]	−0.3 [−0.6, −0.1]	19.3

S25 Second seeds of the variant and of the forgery-head-only model

Table S32: The declared replacement criteria of the variant and the declared comparisons of OURS, forgery head only minus OURS, full model on two seeds, each within its seed (paired video-cluster intervals, 1,000 draws). "Replicated": same verdict, or same interval category, on both seeds.

Criterion	seed 1	seed 2	replicated
(A) symmetric-view variant against OURS, full model, same seed			
(i) whole-crop smoothed genuine FA ≤15% (bil. light/mid/strong)	1.0/1.6/1.7 (three-way 34.9/79.8/90.1)	3.8/5.9/8.4 (three-way 22.0/84.1/96.1)	yes
(ii) Smoothing-90 accused ≤15, above own in [25, 75]	8.1, 30.0 (three-way 8.6, 32.5)	17.8, 46.2 (three-way 12.5, 41.3)	no
(iii) video AUC Celeb-DF-v2 / DFD not lower by >0.010	+0.007 [+0.000, +0.015] / +0.006 [-0.000, +0.012]	+0.007 [-0.002, +0.016] / +0.006 [+0.002, +0.012]	yes
(iv) explanation flip not lower by >1.0 (donor/SD/SDXL)	92.6/92.2/91.9 vs 94.0/93.6/93.5	93.6/93.0/92.6 vs 90.9/90.4/90.2	no
(v) Celeb-DF-B AUC_b not lower by >0.010	0.868 vs 0.866	0.888 vs 0.883	yes
(B) OURS, forgery head only minus OURS, full model, same seed			
Celeb-DF-v2 frame AUC	+0.023 [+0.011, +0.034]	+0.036 [+0.023, +0.048]	yes
Celeb-DF-v2 video AUC	+0.009 [-0.004, +0.024]	+0.017 [+0.003, +0.031]	no
DFD frame AUC	+0.020 [+0.013, +0.028]	+0.006 [-0.000, +0.012]	no
DFD video AUC	+0.009 [+0.003, +0.017]	-0.003 [-0.011, +0.005]	no
Celeb-DF-B AUC_b	+0.030 [+0.014, +0.045]	+0.021 [+0.009, +0.034]	yes
DF40 FR, FF++ domain	+0.124 [+0.113, +0.135]	+0.076 [+0.066, +0.086]	yes
DF40 FR, Celeb-DF domain	+0.145 [+0.133, +0.157]	+0.139 [+0.128, +0.149]	yes

The outcome of the replacement rule replicates (the variant fails on both seeds, so OURS, full model is kept), but not its reason: the first seed fails (iv) and the second fails (ii). The flip-rate cost of the first seed is therefore within seed variation, and the cost on commercial Smoothing-90 is seed-dependent. The intended effect holds on both seeds. The miss rise on whole-crop-smoothed forgeries of the variant is $+9.7$, $+13.8$ and $+11.6$ points on the second seed against $+28.1$, $+33.8$ and $+34.1$ on the first (Table S22); its skin-only false alarms were scored on the first seed only. The advantage of OURS, forgery head only over OURS, full model replicates on Celeb-DF-v2 frames, Celeb-DF-B under the presets and DF40 reenactment, and is seed-dependent on DFD and on Celeb-DF-v2 videos. On the AR sources the second seeds accuse 75.6 and 82.1% (FairBeauty) and 94.3 and 90.3% (B-LFW); at the deployable threshold all three second-seed models accuse 0.1% of Smoothing-90 renders.

S26 DF40: every detector and arm

Table S33: DF40 held-out, untreated. Video AUC (mean frame score per video), mean over methods, with 95% bootstrap intervals over videos. Real videos are the FF++ and Celeb-DF-v2 test reals; the 718 MobileSwap videos from FF++ training identities were removed; 140 fake videos per method in the Celeb-DF domain. FS: nine face-swap methods; FR: twelve reenactment methods. CADDM and the rows from P0 downwards were scored after the first table had been read. Paired blocks: OURS, forgery head only minus each detector with the video and the identity interval of the identity-bootstrap script (Sec. S12; Celeb-DF domain: video only, no identity labels), and OURS, full model minus each detector with the video interval of the earlier version.

	FF++ domain		Celeb-DF domain	
Detector	FS	FR	FS	FR
Xception [2]	0.854 [0.839, 0.869]	0.859 [0.846, 0.872]	0.670 [0.653, 0.688]	0.691 [0.674, 0.706]
EfficientNet-B4 [3]	0.812 [0.796, 0.828]	0.818 [0.804, 0.831]	0.724 [0.706, 0.743]	0.712 [0.697, 0.729]
F3Net [4]	0.826 [0.809, 0.841]	0.797 [0.782, 0.811]	0.697 [0.679, 0.715]	0.671 [0.654, 0.686]
SPSL [5]	0.766 [0.748, 0.783]	0.754 [0.737, 0.770]	0.717 [0.699, 0.739]	0.704 [0.688, 0.721]
UCF [6]	0.843 [0.828, 0.857]	0.825 [0.811, 0.839]	0.716 [0.698, 0.733]	0.704 [0.689, 0.720]
RECCE [7]	0.860 [0.846, 0.874]	0.850 [0.837, 0.863]	0.736 [0.719, 0.753]	0.714 [0.698, 0.729]
CORE [8]	0.850 [0.835, 0.865]	0.858 [0.846, 0.870]	0.656 [0.638, 0.674]	0.676 [0.659, 0.693]
SRM [9]	0.836 [0.820, 0.852]	0.830 [0.816, 0.842]	0.732 [0.715, 0.750]	0.740 [0.725, 0.756]
Effort [10]	0.944 [0.935, 0.953]	0.808 [0.793, 0.822]	0.935 [0.926, 0.944]	0.710 [0.692, 0.726]
Forensics Adapter [11]	0.948 [0.940, 0.956]	0.828 [0.814, 0.841]	0.897 [0.885, 0.908]	0.665 [0.649, 0.681]
GenD [12]	0.966 [0.959, 0.973]	0.935 [0.927, 0.943]	0.947 [0.940, 0.955]	0.828 [0.815, 0.840]
DFD-HR [13]	0.982 [0.977, 0.986]	0.932 [0.924, 0.940]	0.940 [0.931, 0.948]	0.698 [0.681, 0.714]
CADDM [22]	0.869 [0.856, 0.882]	0.794 [0.779, 0.809]	0.788 [0.772, 0.804]	0.541 [0.523, 0.560]
OURS, forgery head only	0.969 [0.962, 0.975]	0.842 [0.828, 0.854]	0.926 [0.916, 0.936]	0.647 [0.630, 0.663]
second seed	0.961 [0.954, 0.968]	0.856 [0.844, 0.868]	0.924 [0.914, 0.933]	0.698 [0.681, 0.713]
OURS, full model	0.952 [0.943, 0.960]	0.718 [0.701, 0.733]	0.905 [0.894, 0.915]	0.502 [0.484, 0.519]
second seed	0.960 [0.952, 0.967]	0.780 [0.764, 0.793]	0.928 [0.918, 0.937]	0.559 [0.540, 0.577]
OURS, full model, sym. view	0.952 [0.944, 0.959]	0.738 [0.722, 0.752]	0.909 [0.898, 0.919]	0.539 [0.521, 0.556]
second seed	0.958 [0.950, 0.965]	0.718 [0.700, 0.734]	0.932 [0.923, 0.941]	0.504 [0.485, 0.522]
FF++-only three-way	0.962 [0.954, 0.969]	0.809 [0.794, 0.823]	0.933 [0.923, 0.942]	0.637 [0.621, 0.653]
P0	0.981 [0.976, 0.985]	0.906 [0.896, 0.915]	0.925 [0.916, 0.935]	0.692 [0.675, 0.707]
P0, seed 2	0.970 [0.964, 0.975]	0.877 [0.865, 0.887]	0.939 [0.930, 0.948]	0.720 [0.704, 0.734]
P0, seed 3	0.979 [0.974, 0.983]	0.873 [0.861, 0.884]	0.931 [0.923, 0.940]	0.633 [0.616, 0.648]
P2	0.942 [0.933, 0.950]	0.809 [0.794, 0.823]	0.907 [0.895, 0.917]	0.644 [0.627, 0.659]
P2, seed 2	0.944 [0.935, 0.952]	0.785 [0.770, 0.799]	0.901 [0.890, 0.912]	0.643 [0.627, 0.658]
P2 no smoothing	0.953 [0.945, 0.960]	0.858 [0.846, 0.870]	0.883 [0.870, 0.894]	0.705 [0.689, 0.720]
P2 smoothing only	0.945 [0.935, 0.953]	0.779 [0.762, 0.794]	0.908 [0.897, 0.919]	0.593 [0.575, 0.610]
P2 smoothing only, seed 2	0.953 [0.945, 0.960]	0.833 [0.818, 0.847]	0.921 [0.911, 0.931]	0.629 [0.611, 0.645]
P1	0.952 [0.944, 0.960]	0.807 [0.792, 0.821]	0.885 [0.874, 0.897]	0.569 [0.551, 0.587]
P1 smoothing only	0.965 [0.958, 0.971]	0.871 [0.859, 0.882]	0.925 [0.916, 0.934]	0.673 [0.656, 0.689]
P1 smoothing only, seed 2	0.963 [0.956, 0.969]	0.869 [0.857, 0.880]	0.940 [0.932, 0.948]	0.689 [0.673, 0.705]
P1 smoothing only, seed 3	0.967 [0.960, 0.973]	0.900 [0.889, 0.910]	0.923 [0.913, 0.932]	0.679 [0.663, 0.695]
P2 symmetric smoothing	0.968 [0.961, 0.974]	0.871 [0.858, 0.882]	0.913 [0.902, 0.923]	0.670 [0.653, 0.686]
P2 symmetric smoothing, seed 2	0.950 [0.941, 0.958]	0.796 [0.780, 0.810]	0.935 [0.926, 0.944]	0.646 [0.628, 0.661]
Effort scheme, symmetric smoothing	0.966 [0.960, 0.973]	0.822 [0.808, 0.835]	0.934 [0.925, 0.943]	0.614 [0.596, 0.631]
released Effort fine-tuned, symmetric smoothing	0.941 [0.931, 0.951]	0.805 [0.790, 0.819]	0.920 [0.910, 0.930]	0.705 [0.688, 0.722]
paired difference, OURS, forgery head only (first seed) minus; video / identity interval				
Effort [10]	+0.024 [+0.016, +0.034] [+0.005, +0.044]	+0.034 [+0.017, +0.051] [-0.005, +0.076]	-0.009 [-0.020, +0.001]	-0.063 [-0.083, -0.044]
Forensics Adapter [11]	+0.021 [+0.012, +0.029] [+0.007, +0.035]	+0.014 [-0.000, +0.028] [-0.022, +0.049]	+0.030 [+0.018, +0.041]	-0.019 [-0.036, +0.000]
GenD [12]	+0.002 [-0.005, +0.009] [-0.011, +0.016]	-0.094 [-0.106, -0.082] [-0.123, -0.064]	-0.021 [-0.031, -0.012]	-0.182 [-0.197, -0.166]
DFD-HR [13]	-0.013 [-0.019, -0.008] [-0.023, -0.004]	-0.090 [-0.103, -0.078] [-0.118, -0.063]	-0.014 [-0.023, -0.005]	-0.051 [-0.066, -0.036]
OURS, full model (same seed)	+0.017 [+0.012, +0.023] [+0.006, +0.027]	+0.124 [+0.113, +0.135] [+0.094, +0.157]	+0.021 [+0.013, +0.029]	+0.145 [+0.133, +0.157]
paired difference, OURS, forgery head only (second seed) minus; video / identity interval				
Effort [10]	+0.017 [+0.009, +0.027] [-0.001, +0.036]	+0.048 [+0.032, +0.063] [+0.011, +0.087]	-0.012 [-0.023, -0.001]	-0.012 [-0.031, +0.006]
Forensics Adapter [11]	+0.014 [+0.005, +0.022] [-0.001, +0.029]	+0.028 [+0.014, +0.041] [-0.006, +0.063]	+0.027 [+0.015, +0.037]	+0.032 [+0.015, +0.049]
GenD [12]	-0.005 [-0.012, +0.002] [-0.018, +0.008]	-0.080 [-0.092, -0.069] [-0.110, -0.051]	-0.024 [-0.033, -0.015]	-0.131 [-0.145, -0.116]
DFD-HR [13]	-0.021 [-0.027, -0.015] [-0.032, -0.010]	-0.076 [-0.089, -0.065] [-0.105, -0.050]	-0.016 [-0.027, -0.007]	-0.000 [-0.014, +0.013]
OURS, full model (same seed)	+0.002 [-0.003, +0.007] [-0.008, +0.013]	+0.076 [+0.066, +0.086] [+0.053, +0.103]	-0.004 [-0.012, +0.004]	+0.139 [+0.128, +0.150]
paired difference, OURS, full model (first seed) minus; video interval				
Effort [10]	+0.007 [-0.002, +0.018]	-0.090 [-0.109, -0.072]	-0.030 [-0.043, -0.018]	-0.208 [-0.228, -0.187]
Forensics Adapter [11]	+0.004 [-0.006, +0.012]	-0.110 [-0.126, -0.094]	+0.009 [-0.006, +0.021]	-0.163 [-0.184, -0.143]
GenD [12]	-0.015 [-0.023, -0.007]	-0.218 [-0.233, -0.203]	-0.042 [-0.053, -0.032]	-0.326 [-0.343, -0.308]
DFD-HR [13]	-0.031 [-0.038, -0.024]	-0.214 [-0.229, -0.199]	-0.035 [-0.046, -0.024]	-0.196 [-0.213, -0.178]

The second seed of OURS, full model is 0.780 and 0.559 on reenactment, that of OURS, forgery head only 0.856 and 0.698. On four Celeb-DF-domain reenactment methods (TPSM, DaNet, MCNet, FaceVid2Vid) OURS, full model scores 0.27–0.37. The three seeds of P0 span 0.09 in the Celeb-DF domain, so differences between arms smaller than that are within the seed spread of the reference there. Four caveats attach to the table. The real videos are the FF++ and Celeb-DF-v2 test reals already used elsewhere in the paper, and Celeb-DF has no identity-disjoint split. Every detector in the table is trained on FF++. Our models were trained on a different crop from the released 256 px DF40 crops; Table S3 gives the size of that handicap on Celeb-DF-v2 and DFD. SBI and LAA-Net are absent because their pipelines need their own crops from full frames.

Reenactment and smoothing across detectors. Over the 13 published detectors scored on both the service and DF40, Spearman correlations of the share of Smoothing-90 renders ranked above their original with reenactment video AUC are

$\rho = 0.36$ ($p = 0.23$; FF++ domain) and $\rho = 0.49$ ($p = 0.09$; Celeb-DF domain); with the share accused at the 5%-FPR threshold, $\rho = 0.62$ ($p = 0.024$) and $\rho = 0.31$ ($p = 0.30$). Holm-adjusted over the four, the smallest is $p = 0.096$. Without CADDM the four are 0.24, 0.35, 0.57 and 0.13. Across our seven FF++-only arms (seed 1) the share above own original and FF++-domain reenactment AUC give $\rho = 0.25$ ($p = 0.59$). P1-smooth ranks 2.6% of renders above their original and keeps reenactment AUC at 0.869–0.900. The data do not support a general link between reading smoothed skin as forgery evidence and detecting reenactment.

S27 DF40 entire-face synthesis

Table S34: DF40 entire-face synthesis, 300 images per method and domain (rng 0): share above the 95th percentile of same-domain DF40 real frames, FF++ / Celeb-DF domain (%). Last row: share that OURS, full model sends to its beautified class by its native decision. Descriptive probe, no bar. Generator stills against H.264 real frames is a confound.

Detector	sd2.1	DiT	SiT	ddim	pixart	StyleGAN3
Effort [10]	99.0 / 87.0	55.0 / 33.3	66.7 / 48.7	65.3 / 42.3	98.7 / 98.3	99.7 / 97.0
Forensics Adapter [11]	76.0 / 49.0	30.0 / 15.0	42.7 / 20.3	25.0 / 8.3	49.0 / 59.0	100.0 / 61.3
GenD [12]	97.7 / 78.0	36.3 / 14.3	46.7 / 23.3	49.7 / 23.3	94.3 / 94.0	99.7 / 96.3
DFD-HR [13]	98.3 / 81.7	43.0 / 11.7	51.0 / 22.0	43.0 / 15.0	98.3 / 93.3	99.7 / 73.0
OURS, forgery head only†	92.0 / 73.0	17.0 / 12.7	17.0 / 15.3	7.7 / 2.7	82.0 / 82.3	99.3 / 50.7
OURS, full model†	72.0 / 56.3	17.7 / 12.3	13.3 / 13.3	5.3 / 2.0	57.7 / 62.7	95.3 / 47.3
OURS, full model, sym. view†	46.0 / 37.7	15.7 / 13.7	13.0 / 14.0	4.7 / 0.3	48.7 / 39.0	55.7 / 5.3
OURS, full model, native three-way decision, share called *beautified* (FF++ / Celeb-DF domain, %)						
beautified	22.0 / 16.3	36.7 / 30.0	47.7 / 41.7	88.0 / 89.0	34.3 / 13.3	3.3 / 19.7

Our models are below Effort and GenD on every method in both domains. By its native decision OURS, full model calls ddim stills beautified in 88–89% of images (forged 7–8%), SiT 42–48%, DiT 30–37% (and real 34–40%); sd2.1, pixart and StyleGAN3 are mostly called forged. The beautified class absorbs synthesised faces with smooth skin, which the class definition calls forged. The variant is the weakest of our models here (StyleGAN3 in the Celeb-DF domain 5.3%).

S28 Fusion of a texture-reading detector with CADDM

Table S35: Score fusion with released weights and no training. Each detector's score is replaced by its quantile among the genuine scores of the same source (for the retouching service: its untouched originals); the fused score is the mean, the minimum or the product of the two quantiles. Video AUC.

	commercial Smoothing-90		untreated video AUC		DF40, FF++ domain		DF40, Celeb-DF domain	
Score	above 5%-FPR thr. (%)	above own orig. (%)	Celeb-DF-v2	DFD	FS	FR	FS	FR
GenD [12] with CADDM [22]								
first member alone	68.7	98.5	0.954	0.977	0.966	0.935	0.947	0.828
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	28.7	92.2	0.970	0.964	0.955	0.942	0.914	0.761
fusion, minimum	23.4	78.2	0.962	0.956	0.925	0.922	0.867	0.708
fusion, product	27.6	87.2	0.969	0.966	0.944	0.935	0.895	0.733
DFD-HR [13] with CADDM [22]								
first member alone	64.9	99.0	0.943	0.976	0.982	0.932	0.940	0.698
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	25.0	91.5	0.963	0.964	0.964	0.942	0.912	0.673
fusion, minimum	21.8	78.1	0.956	0.955	0.932	0.924	0.866	0.659
fusion, product	24.5	87.2	0.962	0.967	0.951	0.935	0.893	0.658
Effort [10] with CADDM [22]								
first member alone	43.2	95.6	0.930	0.972	0.944	0.808	0.935	0.710
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	24.6	86.9	0.965	0.966	0.945	0.881	0.910	0.686
fusion, minimum	22.0	77.4	0.956	0.957	0.915	0.853	0.868	0.658
fusion, product	23.7	84.3	0.963	0.967	0.935	0.869	0.894	0.667
P0 (FF++ only, no beautified views) with CADDM [22]								
first member alone	54.2	98.5	0.938	0.981	0.981	0.906	0.925	0.692
CADDM alone	6.3	60.9	0.930	0.918	0.869	0.794	0.788	0.542
fusion, mean	24.4	91.9	0.961	0.969	0.960	0.939	0.893	0.678
fusion, minimum	20.0	79.1	0.955	0.962	0.929	0.923	0.850	0.665
fusion, product	24.2	87.6	0.961	0.971	0.949	0.933	0.876	0.663

Declared before any fused score was computed: the pairs (GenD, DFD-HR, Effort and P0, each with CADDM), the three rules, and the reading. A fusion counts as useful only if, for the same pair and rule, Smoothing-90 is at most half of the first member's rate, reenactment video AUC is within 0.03 of the first member in both domains, and untreated video AUC is not below the better member by more than 0.01; one rule meeting this for at least two of the three published pairs counts. No rule does (Table S35). Under the mean rule GenD fails the reenactment condition in the Celeb-DF domain (-0.067); DFD-HR and P0 fail the untreated condition on DFD (-0.012 against 0.010); Effort fails the halving condition by 3 points. The minimum rule cuts the accusation further (20.0–23.4%; product 23.7–27.6%), and both lose more reenactment in the Celeb-DF domain (0.66–0.73). The ensemble of the alpha-blending study [26], which combines a blending detector with blend-resistant models, is the closest prior art; our reading differs in that it asks whether fusion changes the ranking of a smoothed face against its own original, and it does not (87–92% of pairs under the mean rule).

S29 Methods that did not meet their bars

Table S36: Every method arm tried under a pre-registered bar. FF++ only unless noted; each block starts with its reference. Bars for the FF++-only arms: Celeb-DF-B frame $\text{AUC}_b \geq 0.890$ and $\Delta\text{MISS}_{\text{beauty}} \leq 7.5$, untreated Celeb-DF-v2 AUC within 0.01 of the reference. None is met. "Random edits" is a random-augmentation control, not an adversarial arm.

	Celeb-DF-B				Celeb-DF-v2	DFD filters
Arm	AUC	AUC_b	$\Delta\text{MISS}_{\text{beauty}}$	FA_b	AUC untreated	AUC_f
random colour grading (reference)	0.938	0.872	+7.5	7.8	0.898	0.932
no grading	0.929	0.849	+14.1	7.2	0.880	0.917
operator, random edits	0.932	0.876	+7.9	7.7	0.873	0.932
operator, worst of 4 draws	0.928	0.867	+12.9	5.6	0.860	0.935
operator, 1-step search	0.930	0.864	+8.2	8.3	0.876	0.935
operator, search within measured edits	0.928	0.869	+15.1	5.2	0.874	0.934
operator, consistency objective	0.924	0.856	+13.0	5.7	0.849	0.922
operator, 3-step search	0.922	0.859	+8.9	7.6	0.849	0.937
decision pooled from the evidence map	0.937	0.877	+14.9	5.0	0.889	0.924
same, no beautified views	0.942	0.891	+14.7	4.4	0.899	0.932
OURS, forgery head only (reference)	0.949	0.896	+7.0	7.4	0.914	0.952
OURS, forgery head only, operator search	0.942	0.892	+9.7	6.0	0.911	0.945
released Effort, mapped (reference)	0.909	0.808	+11.2	4.7	0.886	0.904
released Effort + 2 epochs, no beautified views	0.903	0.805	+5.8	9.4	0.866	0.918
released Effort + 2 epochs, both classes	0.895	0.803	+2.7	10.3	0.861	0.916

The operator. A differentiable edit $B(x; u)$ on a face crop in $[0, 1]$ with every component bounded by a sigmoid: per-channel gamma, gain and lift; a 3×3 colour matrix close to the identity; saturation from monochrome to 1.7; contrast about the mean; a 4×4 luminance field and a per-channel tint field, upsampled bilinearly; a blend toward a Gaussian-smoothed copy; additive grain. Its parameters were fitted by gradient descent to 1,000 pairs each of four measured families (Table S37); the search of the "measured edits" arm starts from a fitted edit and is projected onto the 0.5th–99.5th percentile range of the fitted parameters.

Table S37: The operator fitted to measured edits (training split only). The residual of the commercial renders is geometry, which a photometric operator cannot express.

Measured edit	n	PSNR before	PSNR after fit	10th pct
Tencent retouching	1000	21.4	28.6	24.8
Megvii retouching	1000	29.1	30.6	28.0
training colour filters	1000	21.0	40.1	32.0
random grading	1000	24.2	48.4	39.0

The search works; the training does not carry over. On 400 FF++ validation frames the reference model keeps 96.0% of genuine faces genuine without an edit, 71.0% under a random draw of the operator, 41.0% under the worst of four draws, 8.5% under a one-step signed-gradient search and 0.0% under ten steps; forged faces stay forged in 93.5, 89.5, 83.0, 74.0 and 20.0% of cases. After training on the one-step worst case the genuine accuracy under that search is 71.5% (25.0% under ten steps), and after the three-step variant 76.5% (41.5%). The gain does not carry over to the app presets or to filters from a library the operator was not fitted to, and the forged accuracy under random edits falls from 89.5 to 79.5%: with a label loss, edited views of genuine faces dominate the signal and the model learns that a heavily edited face is less likely forged.

A learned inverse. The exact algebraic inverse of the invertible part of the operator restores an edited frame to 53–58 dB when the true parameters are given. A ResNet-18 estimator trained to predict them from the edited frame alone reaches 26.7 dB on random edits (25.9 before) and 22.7 dB on the held-out filters (21.0 before), and moves untouched frames to 34.4 dB. In front of the detector it changes the false alarms under the held-out filters from 21.5 to 23.6%. A single image does not determine its colour transform.

Edit sensitivity as evidence. The response of the forged logit to six fixed probe edits is larger for genuine than for forged frames on average, but inside the deciles of the clean logit where both classes occur the two respond alike (1.15 vs 0.96; 0.48 vs 0.10), so the asymmetry is a ceiling effect; adding the responses to the logit leaves the false alarms under filters at 16–22%.

A decision pooled from the evidence map. With a zero map target on every beautified item and the forged logit taken as the mean of the top 5% map cells, the map stays dark on filtered genuine faces (mean logit -10.2 against $+1.9$ for the edit map) and the false alarms under filters fall from 21.5 to 14.7%, but the beautification-specific miss rise on Celeb-DF-B grows from 7.5 to 14.9 points.

Fine-tuning a released checkpoint. The released Effort weights were loaded into our pipeline (the mapped model reproduces the released scorer: Celeb-DF-B frame AUC 0.909, AUC_b 0.808) and fine-tuned for two epochs. With both-class exposure the miss rise falls from 11.2 to 2.7 points, the false alarms on beautified genuine faces rise from 4.7 to 10.3%, AUC_b stays at 0.803 and the untreated AUC falls by 0.014; without beautified views the same fine-tuning gives 5.8 points and 9.4%. The operating point moves; separability does not.

S30 The beautified class of the three-way models

The paper reads every model by $P(\text{fake})$ alone. Table S38 gives the native three-way reading of the models that have a beautified class; its numbers are not comparable with the one-rule numbers.

Table S38: Retouching service, native three-way decision: 2,210 FFHQ originals and their renders (four operations at three levels, 26,519 images). Paired rank: share of pairs in which the render gets a higher $P(\text{filter})$ than its own original (50 means no signal).

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, observed forgeries	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Trained with two vendors' renders and their single-operation components; Alibaba never seen (ours)							
Ours, full model (CLIP ViT-L/14 + LoRA)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
Ours, full model, EfficientNet-B4 (browser)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4

Three-way models trained on FF++ alone separate retouched from untouched faces at chance and call commercial smoothing fake, up to 86% of faces at the strongest level. Trained with the single-operation components of two other services, Ours, full model calls 0.6% of the retouched faces fake by its own three-way decision (under the one rule 4.7%), sends 67.7% of them to the beautified class and gives 92.9% of renders a higher filter score than their own original. It also sends 20.8% of the untouched originals to the beautified class, so its balanced accuracy on "retouched or not" is 73.5% (75.4% for the second seed), against a pre-registered target of 75%. The AUROC of $P(\text{filter})$ between originals and renders is 0.80–0.82 for every model trained with components. The symmetric-view variant sends 13.9% of the untouched originals to the beautified class and has a balanced accuracy of 73.4%. On Celeb-DF-B, Ours, full model sends 13.2% of untreated genuine frames, 47.4% of beautified genuine frames, 16.6% of beautified deepfakes and 3.6% of untreated deepfakes to the beautified class (second seed 20.7, 56.9, 16.1 and 3.9%; variant 14.2, 48.1, 17.4 and 3.7%). The attribute recognises about half of the app-beautified genuine frames and labels about one in seven untouched ones as beautified, so it is reported with these error rates and not as a separation result.

Under the one rule the accusation concentrates on smoothing. Table S39 gives the share of renders above the 5%-FPR threshold of the originals per operation.

Table S39: The retouching service per operation under the one rule: share of renders (%) above the threshold that accuses 5% of the untouched originals. Level 90 for the four operations, and the three levels of smoothing.

	level 90			smoothing		
Detector	whitening	face slimming	eye enlarging	level 30	level 60	level 90
Xception [2]	5.7	3.7	4.6	16.6	36.2	51.6
EfficientNet-B4 [3]	5.1	4.0	4.7	11.5	23.9	36.3
F3Net [4]	5.4	4.2	4.4	15.0	33.2	50.6
SPSL [5]	5.2	4.9	4.9	14.9	34.1	51.2
UCF [6]	6.6	4.0	4.2	16.8	32.5	41.4
RECCE [7]	5.3	4.1	4.5	11.9	22.7	36.8
CORE [8]	5.7	4.4	4.8	14.8	36.6	54.8
SRM [9]	4.5	3.8	4.7	18.7	45.0	62.8
SBI [19]	5.4	6.2	5.5	14.8	31.6	37.3
UnivFD‡ [20]	6.2	5.7	11.7	6.8	12.3	16.6
NPR‡ [21]	n/a	n/a	n/a	n/a	n/a	n/a
Effort [10]	11.1	9.6	16.8	13.2	27.2	43.2
Forensics Adapter [11]	5.4	6.2	7.0	10.2	25.8	39.7
GenD [12]	8.1	6.4	7.5	20.7	55.3	68.7
DFD-HR [13]	7.7	8.0	9.6	17.2	46.7	64.9
CADDM [22]	4.7	6.0	5.9	5.4	5.6	6.3
LAA-Net [23]	5.2	4.8	4.8	11.5	17.2	17.2
OURS, forgery head only	5.4	5.0	4.3	5.6	6.3	9.1
OURS, full model	4.8	3.7	3.6	6.2	5.0	8.6
OURS, full model, sym. view	6.3	3.0	3.3	7.3	5.3	8.1
P0	6.8	5.8	5.4	17.0	40.8	54.2
P2-sym	5.8	5.0	5.0	4.5	3.9	5.9

Table S40: All trained variants on the service (2,210 originals, 26,519 renders). Columns as in Table S38.

	originals	retouched (pooled)				paired rank at level 90	smoothing-90
Model	real/fake/filter	→filter	→fake	balanced	paired rank	eye / white / lift / smooth	→fake
FF++ c23 training only (zero-shot)							
three-way, RepViT-M0.9	0.8/2.5/96.7	92.6	6.4	47.9	48.4	73 / 55 / 52 / 15	17.7
three-way, observed forgeries	36.8/5.7/57.5	66.5	9.1	54.5	79.8	97 / 89 / 79 / 59	27.3
three-way, observed and self-blended forgeries	33.8/16.7/49.5	54.6	22.1	52.6	69.2	92 / 83 / 65 / 44	50.2
same, faithful blend	28.2/22.1/49.6	52.8	28.0	51.6	65.7	92 / 77 / 62 / 35	58.4
same, label-symmetric degradation	31.7/26.3/42.0	45.3	33.6	51.7	64.3	89 / 76 / 63 / 35	63.1
three-way, part edits + evidence head	16.7/55.9/27.4	28.6	59.4	50.6	60.9	89 / 78 / 60 / 20	85.8
Variants trained with two services' renders (each read once on the test service)							
renders only (EffNet-B4, s1)	93.4/0.8/5.8	38.1	0.6	66.1	92.8	99 / 85 / 95 / 100	0.1
renders only (RepViT)	92.8/1.4/5.9	36.1	1.1	65.1	92.6	96 / 89 / 93 / 100	0.3
renders + single-operation components (EffNet-B4, s1)	76.8/0.5/22.8	68.2	0.3	72.7	94.2	99 / 90 / 94 / 100	0.4
renders + single-operation components (EffNet-B4, s2)	66.6/0.3/33.2	74.9	0.2	70.8	94.2	99 / 92 / 93 / 99	0.4
renders + single-operation components (RepViT)	43.4/0.9/55.7	83.7	0.5	64.0	95.1	99 / 91 / 96 / 99	1.1
components below noise floor removed (EffNet-B4, s1)	76.7/0.4/22.9	68.8	0.3	72.9	94.9	99 / 92 / 94 / 99	0.5
same + evidence head (EffNet-B4, s1)	83.2/0.6/16.2	62.2	0.5	73.0	94.1	99 / 91 / 93 / 99	0.8
same + part-level forgeries (lite) (EffNet-B4, s1)	81.0/0.4/18.6	63.8	0.3	72.6	93.8	99 / 91 / 93 / 99	0.4
same + evidence head, CLIP (CLIP-L/14, s1)	77.4/0.4/22.2	71.4	0.2	74.6	93.8	98 / 90 / 94 / 100	0.2
same + evidence head, CLIP (CLIP-L/14, s2)	73.0/0.3/26.7	73.6	0.2	73.4	95.1	99 / 94 / 96 / 100	0.2
same + part-level forgeries, CLIP, filter-F1 selection (superseded) (CLIP-L/14, s1)	57.9/0.3/41.9	82.3	0.1	70.2	94.8	98 / 94 / 95 / 100	0.0
same + part-level forgeries, CLIP (CLIP-L/14, s1)	89.5/0.6/9.8	59.2	0.5	74.7	95.5	98 / 94 / 96 / 100	0.1
same + part-level forgeries, CLIP (CLIP-L/14, s2)	83.3/0.4/16.2	67.5	0.2	75.6	93.0	98 / 88 / 95 / 100	0.2
main recipe without colour grading (CLIP-L/14, s1)	82.1/0.6/17.3	65.7	0.3	74.2	92.8	97 / 89 / 93 / 99	0.4
main recipe without service edits on forgeries (CLIP-L/14, s1)	83.8/0.4/15.8	65.6	0.2	74.9	94.7	98 / 94 / 94 / 100	0.1
main data, three-way softmax head (CLIP-L/14, s1)	81.5/0.7/17.8	66.4	0.5	74.3	94.1	98 / 94 / 93 / 99	0.5
OURS, full model (CLIP-L/14, s1)	78.4/0.8/20.8	67.7	0.6	73.5	92.9	98 / 92 / 94 / 99	0.8
OURS, full model (CLIP-L/14, s2)	79.8/0.5/19.6	70.5	0.4	75.4	94.3	98 / 92 / 95 / 99	0.6

S31 Explanations: all variants

Table S41: Restoration test on held-out part-level forgeries (FF++ test frames, test-video donors, detected items only) for every variant. IoU_7: IoU with the exact footprint on a 7×7 grid. Flip: share of items that leave *fake* when the named region, of the footprint's area, is restored to the original. Ceil.: the same for the true footprint. Compl.: the same for the complement. SDXL was never used in training.

	donor transplant				SD-1.5 inpaint				SDXL inpaint (unseen editor)			
Map	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.	IoU_7	flip	ceil.	compl.
centre prior / random region (last col.)	0.310	38.6	–	17.0	0.249	40.9	–	17.9	0.255	40.2	–	17.7
no head, Grad-CAM [27]	0.367	50.9	85.8	15.2	0.259	45.4	85.2	25.0	0.269	48.0	85.9	23.6
no head, Grad-CAM++ [28]	0.414	51.7	85.8	7.2	0.327	47.8	85.2	16.7	0.330	48.8	85.9	19.1
no head, Score-CAM [29]	0.408	56.5	85.8	5.9	0.334	52.4	85.2	15.4	0.341	52.4	85.9	17.1
head, part edits, FF++ only	0.723	84.0	84.7	0.4	0.752	82.6	83.2	0.7	0.746	82.9	83.8	0.7
+ symmetric degradation	0.723	86.3	87.1	0.4	0.719	82.5	83.7	2.0	0.728	83.7	84.4	1.5
+ restoration-consistency loss	0.759	89.4	90.0	0.4	0.614	75.2	79.8	4.4	0.684	82.5	84.6	2.6
+ consistency + symmetric degradation	0.722	88.4	89.3	0.7	0.723	85.2	86.1	2.4	0.732	86.1	86.9	1.8
head, vendor data, no part edits (EffNet-B4)	0.418	61.4	82.4	12.0	0.280	39.5	63.5	23.4	0.315	46.2	72.3	22.5
head, vendor data, no part edits (CLIP)	0.479	78.8	89.4	2.5	0.376	53.9	69.1	14.4	0.445	68.9	78.3	8.4
Ours, full model, EffNet-B4 (browser)	0.807	87.9	88.0	0.0	0.820	87.4	87.6	0.4	0.812	86.9	87.2	0.4
full-data CLIP model, filter-F1 selection (superseded)	0.832	91.7	91.9	0.1	0.845	91.5	91.6	0.2	0.827	90.9	91.2	0.5
Ours, full model (CLIP ViT-L/14 + LoRA)	0.834	94.0	94.1	0.1	0.841	93.6	93.7	0.4	0.831	93.5	93.6	0.3

The donor-transplant test set uses donors from test videos only. A first build of that set drew donors from training videos (1,023 donor-frame and part pairs occurred in training); it was rebuilt, and the numbers of Ours, full model moved by less than 0.2 points. Figure S5 shows examples. The test is a localisation test plus a causal check that restoring the localised region removes the detection; it does not compare maps across models, evaluate missed forgeries or false accusations of genuine faces, or use a non-oracle infill, and these controls are left for future work.

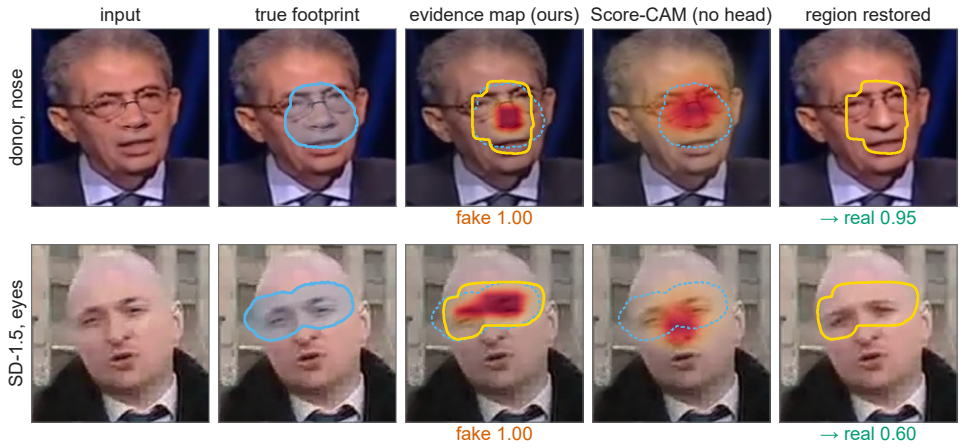


Figure S5: Held-out part-level forgeries: input, exact footprint (blue; dashed on the maps), evidence map with the three-way decision, Score-CAM on the same backbone without the head, and the result of restoring the region cut from the evidence map (yellow, same area as the footprint) to the original, with the decision on it.

S32 Sources that were examined and excluded

AR beauty filters as training data. A model trained with paired AR-filter images called 94.8% of the filtered images *filter* with the face hull greyed out: the capture pipeline, not the face, carries the label, so AR pairs are not used for training. As a test source they are scored in Sec. S6, with a re-compression control and the stated upper-bound reading. **WildDeepfake.** The 800 test frames available to us come from six sequences, which gives intervals of 0.3–0.5 AUC; not reported. **DFDC, DFDCP.** Not available to us in a form with genuine and forged frames from the same pipeline.

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